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The New Brethren of Purity: Ibn Turka and the Renaissance of Pythagorean Science in the Early Modern Persian Cosmopolis

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Abstract

The Brethren of Purity are rightly acclaimed the most influential Pythagorean thinkers of pre-Mongol Islam; their *Epistles* were crucial to the development of a wide and weird range of western and eastern forms of occultism, Sufism and Neoplatonic philosophy, as current research has shown. What has yet to be mapped is the extent to which their post-Mongol Persianate heirs – the self-styled New Brethren of Purity – exerted an even wider and weirder sociopolitical effect. From metaphysics to physics, empire to empiricism, painting to architecture, astronomy to historiography, poetry to magic, their radical reformulations of lettrism (*‘ilm al-ḥurūf*) as universal science and theurgic technology became a primary driver of early modern Pythagoreanism and Platonism across the Afro-Eurasian ecumene – indeed a key to the category of early modernity itself. (And perhaps postmodernity too.) This article sketches the outlines of such a map by charting the legacy of Ibn Turka of Isfahan (d. 1432), foremost member of the New Brethren in the Timurid east and their philosophical powerhouse, who was most responsible for rewriting lettrism as Pythagoreanism and Pythagoreanism as *mathesis* and *mathesis* as total transhumanist empire. A Muslim Iamblichus and Cusanus and Bruno and Pico and Dee combined, Doctor Littleturk's example enforces a larger, less doggedly Eurocentric and classicocentric framework for investigating how the mathematizing Greater Western tradition, past and present, might be written and might write otherwise.

Keywords

Persian Pythagoreanism – occultism – lettrism – Brethren of Purity – New Brethren – Ibn Turka – science and empire – early modernity – transhumanism – *mathesis*

The often Pythagorean-*cum*-kabbalistic tenor of early modern Latin Christian intellection is now widely recognized, and scholarship on the major exponents of this current, from Nicholas of Cusa and Giovanni Pico della Mirandola to John Dee and Johannes Kepler, has burgeoned in recent decades. Spurred by Frances Yates's (d. 1981) Brunonian vision, some revisionist historians have even posited the fifteenth to seventeenth-century florescence of Pythagorean-kabbalist thought as an important matrix for the so-called Scientific Revolution – essentially a *mathematical* revolution – and hence scientific modernity as such.¹ That the intellectual and imperial history of the contemporary Islamic and especially Persianate world was at least equally so, however, has barely begun to be pursued by specialists. Not only does the golden age of Islamic Pythagoreanism remain almost entirely unstudied; its sweeping implications for how we write Western, Afro-Eurasian and indeed modern colonial history more broadly have yet to be grasped.²

Such a gross imbalance in the literature, patently colonialist-Orientalist in origin, continues to be driven by the reflexive Euro-, Arabo- and classicocentrism of Western intellectual history generally and Islamic specifically, whereby post-twelfth-century developments in the latter – when Latin Christian interest in and knowledge of Arabic scholarship effectively ceases – may be safely amputated as irrelevant afterthought, as somehow Not-Western.³ Thus we are permitted to speak, for example, of the Ikhwān al-Ṣafā' of tenth-century Iraq

1 While the Yates Thesis, in its extreme form, is now generally rejected, her primary insight still holds, and has proved especially productive for (Europeanist) history of science and philosophy. See e.g. Killeen and Forshaw (eds), *The Word and the World*; Håkansson, *Seeing the Word*; Hanegraaff, "Beyond the Yates Paradigm."

2 Given the total exclusion of its hugely influential Persianate iterations to date, I here use simply "Pythagoreanism" rather than "Neopythagoreanism," specific to late antiquity, since "Neo-neopythagoreanism" to describe the exciting new post-Mongol developments summarized below is unusuabily awkward, so would only further ghettoize this landmark era in Western thought.

3 For intellectual-historical purposes, the early modern West is best defined as *the half of Afro-Eurasia* incorporating the *Arabic, Persian, Greek and Latin cosmopolises*, that vast realm where the *Hellenic-Abrahamic dialectic* reigned supreme, and science and empire came to be pursued in *mathematical-linguistic* terms; see Melvin-Koushki, "*Tahqīq* vs. *Taqlīd*"; idem, "Translating Esotericism."

as culmination of late-antique occult-scientific Pythagoreanism;⁴ some edited collections in the field of (Neo)pythagorean studies feature a chapter (usually token, to be sure) on this anonymous confraternity of ‘Abbasid encyclopedists.⁵ Even the most recent (2022) collection – which claims, falsely, to give “equal weight to the Arabic, Hebrew and Persian cultures” – likewise insists that Islamic intellection is a strictly medieval and mostly Arabic phenomenon, and “Renaissance” and “early modernity” can only be Latin European categories.⁶ The simple fact that Islamic cultures encompassed a full two-thirds of the human race by the sixteenth century and constellated both the Afro-Eurasian ecumene and the bulk of Western intellection remains unimaginable to academic orthodoxy.

Whence this dogged refusal by modern historians to acknowledge the existence, profusely documented, of a wonderfully robust early modern Persianate Pythagorean school of thought and action?

Enter the New Brethren of Purity, who were far more intellectually and imperially influential than their tenth-century namesake.⁷ This network of equally ardently Pythagoreanizing occult scientists radiated from Cairo north to Anatolia and east to Iran, Transoxania and India, and featured celebrated Arabo-Persian scholars whose projects are both contemporary and directly

4 Fowden, *Before and after Muḥammad*, 204–7. The association of (Neo)pythagoreanism with occult science – especially of *Persianate* vintage – is of very long standing, with late antique authors such as Pliny the Elder (d. 79 CE) and the Neopythagorean Numenius of Apamea (fl. mid-2nd c.) presenting Pythagoras as a disciple of Zoroaster or the Persian Magi in natural magic. See Huffman, “Pythagoreanism,” *SEP*, §4.5; Joost-Gaugier, *Measuring Heaven*, pp. 40–41. This relationship, however, was sometimes reversed, with the dualist Zoroaster as a deficient disciple of the monist Pythagoras (De Smet, “Les philosophes grecs,” p. 828).

5 That medieval and early modern Latinate developments are likewise given short shrift in most surveys on this theme – though naturally not nearly to the same astonishing extent – is symptomatic of the same classicocentric ill; representative here is Carl Huffman’s otherwise excellent recent edited collection *A History of Pythagoreanism*, whose 21 chapters include two on the Middle Ages and early Renaissance respectively, seemingly by way of a postscript.

6 Caiazzo, Macris and Robert (eds), *Reception of Pythagoras*, p. 3. Full disclosure: the present article was an invited chapter in this volume; submitted in 2018, it passed extensive peer review and was set to print – then in 2021 cut from the volume by the series editor, Kyriakos Demetriou, on bizarrely nationalistic grounds, and without notice. My experience is of course anecdotal; but the fact that it took a full seven years for this, the first article to introduce early modern Persian Pythagoreanism, to be published, emblemizes this strange scholarly unknowing in the face of an astonishing bonanza of surviving archival and artifactual evidence.

7 On the pregnant self-identification of this early fifteenth-century loose scholarly network as *ikhwān al-ṣafā’ wa-khullān al-wafa’* see Binbaş, *Intellectual Networks*, pp. 106–13.

comparable to those of Cusanus, Pico, Bruno, Dee and Kepler – but who won a vaster, and more militarily potent, Western readership.

Recent research on the New Brethren has begun to map their fifteenth-century sociological and religiopolitical contexts, identifying lines of inquiry useful to specialists. But it too has weirdly failed to engage their self-styling as *Pythagoreans*, much less their place in the history of Western philosophy; the explicit, imperial and indeed monumental persistence of their Pythagorean project through at least the seventeenth century, with resonances through the nineteenth (think the Tāj Maḥall as starship); and their equal importance to the history of science and technology (think the Samarkand Observatory also as starship) and a wide range of other theoretical and practical arts throughout the Greater West in its many kaleidoscopic runups to modernity. These failures to engage leave the New Brethren – among the most influential (trans)humanists of the early modern era, and iconic champions of the Imperial Cosmic Algorithm – an object of boutique Orientalist medievalizing interest only, rather than crucial conversation partners who might help us understand and navigate our own Algorithmic-Transhumanist-Starstraining moment, to which they helped give rise.⁸

As a call to begin to right these bizarre wrongs in the field of Pythagorean studies, this article briefly introduces the New Brethren of Purity and their legacy to the present. My overview below, it must be emphasized, is strictly preliminary and suggestive; many decades of basic philological spadework will be necessary to grasp the general contours and impact of their project, to say nothing of its textual details. (Almost all their relevant writings remain in

8 Binbaş's *Intellectual Networks* – the first and only book-length study of the New Brethren to date – is unfortunately a case in point. Preoccupied with the medievalness and hence curious irrationality of its actors as religiopolitical beings only, the study is crippled by painfully incorrect, oddly truncated and blatantly tendentious translations from both Persian and Arabic, and paradoxically cleaves to an “Enlightenment” western European theme to show the New Brethren to be anything but Enlightened – certainly of no relevance to the history of Pythagoreanism, much less history of science. Two examples taken at random must here suffice: its account of Ibn Turka's radically Hermetic-theurgic Persian treatise *Splitting of the Moon and the Last Hour* (pp. 160–62) is not just founded on simple mistranslation, but mistranslation that erases the philosophical, scientific, technological and indeed (trans)humanistic stakes clearly set forth; and its account of the infamous anti-Ottoman revolutionary Shaykh Badr al-Dīn (Bedreddīn) and his Arabic work *Inspirations* (pp. 122–32) centers on a mistranslation so bizarrely semiocidal as to cast doubt on the interpretive framework of Binbaş's book as a whole. (For a correct translation, see Melvin-Koushki, “Toward a Neopythagorean Historiography,” p. 397 n. 44.) To be fair, my own 2012 dissertation, taking Binbaş's as its point of departure, missed many of these stakes and connections too; but I hope to have since made amends in the studies cited and summarized below.

manuscript, a majority of them inaccessible or being destroyed as we speak.) But grasp it we must. For their example is not merely instructive in its own right, but indispensable to any comparative Western and global intellectual and imperial history worth the name: it constitutes an index of the emergence of a distinctive Western early modernity, and so continues to define the parameters of what it means to be modern and postmodern too.

The New Brethren, as architects of one of the greatest flowerings of Pythagoreanism in Greater Western history, were naturally perennialist, hellenophile champions of the *prisca theologia* in precisely the same fashion as both their Old Brethren predecessors and their Latin Christian Renaissance peers. Unlike either party, however, they resolved the Quarrel of the Ancients and the Moderns in favor of the Moderns, which is to say, themselves. Where Renaissance humanist scholarship is insularist, neoclassicizing and declinist, demanding a return to the “purely” Greek golden age of antiquity, that of the New Brethren is cosmopolitan, progressivist and supersessionist: the golden age is now.⁹

It is also imperially universalist. Neither the Old Brethren nor Renaissance mages could plausibly dream of universal Alexandrian empire, Islamic or Christian; but the New Brethren – alacritous students and theorizers of Mongol pan-imperialism – ably served the great early modern Turko-Mongol Perso-Islamic empires as political philosophers, ideologues and indeed shamans in pursuit of that Skystraining dream. So successful were their efforts, moreover, that post-Mongol Islamic imperialism became more explicitly Pythagorean and Hermetic-theurgic in tenor than any other form thereof in Western history, either before or since. Aspiring Muslim *saint-philosopher-kings*, on the model of Archytas but of Alexandrian ambition and vision, multiplied across the Persian cosmopolis and through the seventeenth century together dominated a full third of the Afro-Eurasian ecumene.¹⁰ By contrast, Dee’s remarkably similar call in the late sixteenth for a bihemispheric “Brytish Impire” was abortive, because premature;¹¹ and the European colonialism of later centuries, needless to say, was not particularly Pythagorean, and violently anti-Alexandrian.

9 So I argue in Melvin-Koushki, “*Tahqīq* vs. *Taqlīd*.”

10 See Huffman, *Archytas of Tarentum*; Melvin-Koushki, “Early Modern Islamicate Empire.” On modern scholarly resistance to discussing even ancient Pythagoreanism in political terms, see Cornelli, *In Search of Pythagoreanism*, pp. 23–33.

11 This the Welsh Pythagoreanizing occultist and imperial ideologue did for Queen Elizabeth I (r. 1558–1603) in his *Britanici imperii limites* of 1593. For a counterfactual account of Dee’s occult-scientific imperialist project vis-à-vis its (far more successful) early modern Islamic cognates, see Melvin-Koushki, “Dr. Dee’s Ottoman Adventure.”

As for science, occult or otherwise: the New Brethren, quite simply, would seem to have been the first Western scholars to successfully institutionally call for the *mathematization of the cosmos* – a project that has defined much of Western intellection from Greek antiquity to Euro-American modernity. That modern physics or hegemonic digital culture or AI itself does not claim the handle Pythagorean is more an issue of branding (aka magic) than proof of epistemic rupture.¹²

In short: The West European developments Western intellectual historians have long associated with the emergence of early modernity were preceded, then paralleled, by precisely cognate developments in the Persianate east, culturally genetically equally the West; and the latter were driven by the New Brethren of Purity and their self-named heirs through at least the seventeenth century, with significant technoscientific and religiocultural reverberations to the present. (Computing is among the most notable modern instances of the first, the Bahā'ī Faith of the second.) To cease historiographically explaining that early modernity, and hence (post)modernity, in the crassest of Eurocentric and teleological terms, we must finally study these, the Cusanuses, Picos, Brunos, Dees and Keplers of Islam.

Ibn Turka and the New Brethren of Purity

As entrée to the project of the New Brethren more broadly, I here present the most important member of this scholarly confraternity in Timurid Iran: Ibn Turka of Isfahan (1369–1432). This occult philosopher's project centered on the reformulation and philosophical systematization of lettrism (*'ilm al-ḥurūf*) as universal Pythagorean science and Iamblichean theurgic technology – and hence basis for Mongol-Islamic universal empire.¹³ His *magnum opus*, the

12 Melvin-Koushki, "Of Islamic Grammatology"; cf. Kahn, *Pythagoras and the Pythagoreans*, pp. 171–72.

13 Among the occult sciences that became permanently intertwined with Islamic culture from its very inception, particularly in early Shi'i circles, it is letter science, or lettrism, that underwent the most complex evolution. Most significantly, it eventually emerged as the most *Islamic* of all the occult sciences despite its explicitly late antique Hellenistic, non-Islamic parentage – this due to its jazz-philosophical and jazz-scientific reformulations in the early thirteenth century by Ibn al-'Arabī, Sa'd al-Dīn Ḥamūya and Aḥmad al-Būnī in particular, codified and systematized by the New Brethren. Conceptually, as an umbrella science, lettrism encompasses and integrates the full suite of "magical" and "divinatory" technological applications. Letter-magical techniques include most prominently the construction of talismans (sg. *ṭilasm*), whose engine is usually a magic square (*wafq al-a'dād*), to be populated with letters or numbers relevant to the operation at hand; these

Investigations (*Kitāb al-Mafāḥiṣ*), is best described as the first summa of Islamic Pythagoreanism; it was cited as such for centuries after.¹⁴

For all that this self-styled *Pythagoras redivivus* has been almost totally elided in modern scholarship, and most of his lettrist works – including the *Investigations* – left unpublished and unstudied, Ibn Turka's efforts were not in vain.¹⁵ To the contrary, his oeuvre marks an intellectual and cultural watershed on five major fronts, briefly surveyed in the balance of this article, after a sketch of the New Brethren network as a whole:

- 1) *Metaphysics*: It (re)established occult-scientific Pythagoreanism as a mainstream current in Islamic philosophy down to the modern era, and especially in post-Mongol Persianate cultures, defined, Renaissance-style, by a dramatic philhellenic-antiquarian turn, when the *Epistles* of the Old Brethren again came into vogue after centuries of neglect – and philosophy again *was lived as magic*. Or rather, *Iamblichean-Hermetic theurgy*:

are designed to harness the specific letter-numerical virtues of personal names, whether of humans, jinn, angels or planets; key phrases or quranic passages; and/or one of more of the names of God. (The latter operation is a typical example of the Sufi-occultist practice of 'assuming the attributes of God' or *theomimesis* (*takhalluq bi-akhlāq Allāh*) – hence the divine names as a major focus of lettrism, often termed for that reason '*ilm al-ḥurūf wa-l-asmā*', 'the science of letters and names,' or even simply '*ilm al-asmā*', 'the science of names.'). Letter divination, for its part, includes most prominently the construction of a Comprehensive Prognosticon (*jafr jāmi*'), a 784-page text containing every possible permutation of the letters of the Arabic alphabet. From such a handheld Cosmic Computer may be derived the name of every thing or being that has ever existed or will ever exist, every name of God in every language, and the knowledge of past, present and future events – especially political events – to the end of time. This divinatory aspect of lettrism is associated in the first place with both the House of the Prophet and the the mysterious separated sura-initial letters in the Quran (*muqatta'āt*), similarly held to contain comprehensive predictive power, and to have inspired the basic lettrist technique of *taksīr*, separating the letters of words or names for the purposes of permutation (cf. the sister kabbalist techniques of *gematria* and *temurah*). Most letter-magical and letter-divinatory operations are profoundly astrological in orientation, moreover; careful attention to celestial configurations is essential for the success of any operation, and letter magic often involves the harnessing of planetary spirits (*taskhīr al-kawākib*), together with angels and jinn. Fasting, a vegetarian diet, seclusion and maintenance of a state of ritual purity – shades of originary Pythagoreanism – are also regularly identified as conditions of practice in manuals on these subjects. By the early modern period, *jafr* came to be used in Persian metonymically for lettrism as a whole.

- 14 To be clear, the purpose of this article is not to introduce Ibn Turka's thought, which I have done in the studies cited throughout, but rather to chart his reception and legacy as historical and conceptual framework for the comparative study of the New Brethren, Islamic Pythagoreanism and *mathesis* more broadly going forward.
- 15 My editions and translations of his lettrist works are almost complete; ten are forthcoming in Melvin-Koushki, *Persian Pythagoreanism*.

Ibn Turka is the first thinker in the Greater Western (Helleno-Judeo-Christo-Islamic) tradition to develop a specifically Pythagorean-Platonic mathematical-theurgic Selenocentric philosophy that could scientifically, politically and culturally challenge the hegemonic Avicennan Aristotelian-Platonic hybrid system, turning its sensory theory quite literally, or rather *lettrarily*, on its head.¹⁶

- 2) *Physics*: Beyond driving an occultist renaissance in the natural sciences, it helped inspire the celebrated *mathematical turn* in Arabic astronomy (aka “real science”) in the early fifteenth century – the immediate precedent and parallel, historians of science are now arguing, for the Copernican revolution slightly to the west, and hence crucial to any history of the emergence of scientific modernity. In particular, its successful and unprecedented push for arcminute precision in the *New Sultanic Star Tables* achieved by his friends at the Samarkand Observatory is the epitome of *mathesis*, and would inspire the new theory of *metatime*.
- 3) *Empire and history*: It is responsible for the Pythagoreanization of Timurid imperialism, achieved by means of a new type of *imperial grimoire*, whereby ambitious Muslim sovereigns could scientifically style themselves *saint-philosopher-kings*; this served in turn as model for the great Islamic empires of the early modern era. (Here such famed ancient Pythagorean sovereigns as Archytas of Tarentum (Ar. Arkhūṭās, d. 347 BCE), Greek mathematician-general extraordinaire and likely model for Plato’s own theory of philosopher-kingship, were explicitly or implicitly taken as exemplars.¹⁷) It also inspired a form of *Pythagorean historiography* unprecedented in the Western tradition, whereby history was reframed as an astro-lettrist discipline and so by definition scientific.
- 4) *Book, visual and architectural culture*: It formulates the first Islamic philosophy of language to valorize *textuality* and *visuality* over orality, thus presaging the early modern boom in psychedelic Persianate book and visual culture (including the new genre of album preface), as well as providing theoretical impetus for the poeticization of everyday life so definitive of post-Mongol literary and architectural practice: the world as

16 The celebrated Syrian Pythagorean-Platonist philosopher Iamblichus (d. 325 CE) – whose disciples were responsible for designing the Hagia Sophia, an architectural and occult-technological feat with which the Taj Mahall quite consciously rhymes – here provides the most salient late-antique precedent for Ibn Turka’s own post-Avicennan imperial system, pivoting on *godwork* (theurgy as theosis) and planetary ascent by means of ritual mathematical language. See Adams, *Hagia Sophia*; Shaw, *Hellenic Tantra*.

17 On whom see Huffman, *Archytas of Tarentum*.

linguistic-mathematical transhumanizing *riddle* – and the mosque and mausoleum as *starship*.

- 5) *Reincarnationism and vegetarianism*: While not explicitly *reincarnationist*, unlike that of the Old Brethren or those of their Jewish (but not Christian) cognates, the New Brethren's Pythagoreanizing project nevertheless heralds the revival of reincarnationist doctrine across the early modern Persianate world, epitomized by the Mughal (Indo-Timurid) imperial embrace of the same. Similarly, post-Ibn Turkian Persian grimoire culture also requires of its adepts *vegetarianism* to ensure the effectiveness of astro-lettrist and other occult-scientific operations, especially of the theurgic kind.

Ibn Turka's project, in sum, is immediately cognate to those of his Renaissance peers Cusanus and Pico, but precedes theirs by decades; it further inspired the mathematization of astronomy that would so inspire Kepler, and heralds Dee's kabbalist-imperialist project of the next century. Its ensorcelling literary flavor is likewise Brunonian.¹⁸ But in terms of raw intellectual, cultural and especially imperial impact, it was considerably more successful than theirs; and for basic structural reasons, it did not lead to a soi-disant Enlightenment, that is, *to us*. Disenchantment is a tall tale peculiar to colonialist Europe.¹⁹

This divergence notwithstanding, the projects of these Western worthies together represent a remarkable convergence. For just as Jewish-*cum*-Christian *kabbalah* emerged as the primary expression and marker of Pythagoreanism in Renaissance Europe,²⁰ so too did *lettrism* in the vaster, wealthier and more cosmopolitan contemporary Islamic world; and the latter-day disciples of Pythagoras dominated and shaped the philosophical, scientific, technological, political and aesthetic discourses of Afro-Eurasian Western early modernity as a Helleno-Judeo-Christo-Islamic whole.

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The Mongol conquest of Asia in the mid-thirteenth century and the Black Death in the mid-fourteenth made for a combined apocalypse; in their rupturous wake Muslim scholarly and ruling elites increasingly seized upon Occult

18 De León-Jones, *Giordano Bruno and the Kabbalah*.

19 Josephson-Storm, *The Myth of Disenchantment*.

20 This connection is made explicit by Pico and his fellow Christian kabbalist-perennialist Johannes Reuchlin (d. 1522), who asserted "the philosophy of Pythagoras took its beginning from the precepts of the cabalists" (trans. in Heninger Jr., *Touches of Sweet Harmony*, p. 245).

Science as salvation – and imperial weapon. Engineered by the New Brethren of Purity and their broader knowledge community, the occult-scientific renaissance that swept the Islamic heartlands by the turn of the fifteenth century – like that other, equally occult-scientific Renaissance slightly to the west a few decades later – remained a mainstream cultural force through at least the seventeenth, in places even the twentieth.

The pivot of this highly mobile network of scholar-occultists, the OG Lettrist International,²¹ was Sayyid Ḥusayn Akhlāṭī (d. 1397), Tabrizi Kurdish alchemist, lettrist and geomancer, personal physician to the Mamluk sultan al-Malik al-Ẓāhir Barqūq (r. 1382–99) and rumored messiah. While Akhlāṭī's own oeuvre is scattered, unsystematic and exclusively practical in tenor, he was teacher to the greatest occult philosophers of the next generation, who first systematized lettrism as universal-imperial science. They include in the first place Ibn Turka, his deputy (*khalīfa*) to the Timurid east; his deputy to the Ottoman north was the revolutionary Shaykh Badr al-Dīn of Simavna (executed 1420), who rather than support empire challenged it: he inspired the most flamboyant rebellion in early Ottoman history with his platform of millenarian Ibn 'Arabian monism, to be expressed religiopolitically, economically and militarily in the form of pure communism and monotheistic communion among the impoverished Muslim and Christian peasantry of Rumelia and armed resistance to oppressive Ottoman elites.²² Badr al-Dīn was an associate in turn of 'Abd al-Raḥmān al-Biṣṭāmī of Antioch (d. 1454), whose voluminous works on the occult sciences generally and lettrism specifically would come to serve as important prop to nascent Ottoman eschatological-imperial ideology and political technology, just as Ibn Turka's more streamlined works did for Timurid.²³

21 In current Europeanist scholarship, the term lettrism (unlike kabbalah) is associated almost exclusively with the modern French surrealist movement of the same name, founded by Isidore Isou (d. 2007) in Paris in the mid-1940s; a schismatic branch of this movement termed itself the Lettrist International. While an important aspect of twentieth-century French avant-garde culture, lettrism as a mainstream Western (Islam-Judeo-Christianate) phenomenon, like its sister science kabbalah, is of vastly greater historical significance and influence than is implied by its modern, very limited reception in France. Also like kabbalah, "lettrism" is a first-order term, at least in its translated adjectival form: the New Brethren routinely identify themselves as *ḥarfīs* or *ahl al-ḥarf*, or literally "lettrists."

22 This easily romanticized figure has inspired a rash of modern scholarship, which has tended, however, to elide his immediate intellectual context; the allure of religionism remains considerable. See e.g. Balivet, *Islam mystique*.

23 The first monograph-length study of the New Brethren as an informal network is Binbaş, *Intellectual Networks*; see also Fleischer, "Ancient Wisdom"; Fazlıoğlu, "İhvanu's-Safâ ve

Bisṭāmī thus stands as Ibn Turka's Anatolian cognate; but where the former was encyclopedic in his approach, compiling masses of material from hundreds of authorities (most of them Sufi),²⁴ the latter was philosophically rigorously systematizing and synthetic, and cites few authorities apart from Ibn al-'Arabī and Sa'd al-Dīn Ḥamūya, jazz philosophers extraordinaire.²⁵ Where Bisṭāmī's lettrism is primarily Būnian, that is, focused more on praxis, Ibn Turka's is primarily Ibn 'Arabian-Ḥamūyan, focused more on theory; and where the first is open to Sufism as a parallel path to (trans)human perfection, the latter pointedly reformulates lettrism as a mode of theory and praxis not just distinct from but superior to Sufism, proper to spiritual prepubescents.²⁶ His openly revolutionary Pythagorean project: to establish lettrism, *contra* both Avicennan Peripateticism and Suhrawardian Enlightenment, as the only philosophically valid form of metaphysics-physics – and a primary theoretical and practical basis of Mongol-Islamic universal, mathematical empire. To this end, Ibn Turka cultivated a series of Timurid rulers as occult philosopher-kings and put forward a simultaneously mathematizing and imperialist vision.

Unlike the Old Brethren, moreover, Ibn Turka and his lettrist confraternity did not labor in anonymity, but rather constituted a loose intellectual network comprising some of the most prominent men of letters and imperial ideologues of the era and spanning the Mamluk, Timurid and Ottoman realms via Cairo, its pivot. Nor did the cachet of that network soon fade; likeminded scholars routinely identified with its members for centuries after. Evidence suggests, moreover, that even Byzantine scholars were implicated; the great Neoplatonist Gemistos Plethon (d. 1452) himself – whose own revolutionary project, its “paganizing” thrust aside, is strikingly consonant with that of the New Brethren – may well have met some of their number or learned of their exploits during his sojourn in Ottoman territory.²⁷

Abdurrahman Bisṭāmī”; Melvin-Koushki, “The Quest for a Universal Science,” pp. 16–19 and *passim*.

24 See Gardiner, “Occultist Encyclopedism.”

25 Uy, “Riffing on/as the Real”; *idem*; *Lost in a Sea of Letters*. See also Melvin-Koushki, “Lettrism, Astrology and Jazz Philosophy.”

26 Melvin-Koushki, “Better than Sufi Sex.” On the celebrated sufi-mage Aḥmad al-Būnī (d. btw. 1225–33) and his seminal lettrist legacy see Gardiner, “Esotericism in a Manuscript Culture.”

27 Siniosoglou, “Sect and Utopia.” More generally, Byzantine scholars remained interested in occult-scientific developments in the contemporary Islamicate world long after their Latin peers had ceased to be, and even translated Persian treatises on geomancy (*‘ilm al-raml* > Gr. *rabolion*, *ramplion*), for example, on which Pythagorean science see below.

But the Brethren did not work unopposed. Indeed, the best index of the sweeping success of their consciously revolutionary project is precisely the howls of protest it elicited from puritanical scholars, who explicitly damned the resurgence of Pythagorean-Platonic (*fīthāghūrī-aflātūnī*) occult philosophy then beginning as the greatest threat to (Sunni) Islam in the post-Mongol era, together with the continued mainstream status of Avicennan Peripateticism as a parallel current. These anti-occultism polemicists include in the first place the Damascene jurist-theologian Ibn Qayyim al-Jawziyya (d. 1350) and, now more famously, the historian-jurist Ibn Khaldūn (d. 1406). Railing at length against the occult sciences on rationalist and traditionalist grounds alike, the first claimed the contemporary scholarly embrace of heretical philosophizing to be both a mere repackaging of Isma'ili esotericism and the direct spiritual cause of the catastrophic Mongol conquest. The second went further, and in his status as a Mamluk official sought to have philosophy and the natural-mathematical occult sciences outlawed by state fiat. More: Ibn Khaldūn, a Maliki judge, attempted to have Ibn Turka and his fellow Brethren in Cairo executed as sorcerers. His grudge was personal.²⁸

28 On Ibn Khaldūn's rivalry with the New Brethren in Cairo, see Gardiner, *Ibn Khaldūn versus the Occultists*; Melvin-Koushki, "In Defense of Geomancy"; idem, "Imperial Talismanic Love." Maliki jurists, it should be noted, were notorious as hanging judges in the Mamluk Sultanate (Wiederhold, "Some Remarks").

This is not to say, however, that such polemics were entirely without basis. A certain interest in the occult sciences generally and in what might be called "proto-lettrism" specifically was certainly shown by some earlier Isma'ili authors, though in strictly sporadic and unsystematic fashion (my thanks to Alex Matthews for this observation). A prime example is Abū Ḥātim al-Rāzī (d. 934), who briefly proposed a letter cosmogony reminiscent of the perhaps contemporary *Sefer Yetzirah*, though one strictly limited to the phonology of the primordial letters as roots of creation (Acevedo, *Alphanumeric Cosmology*, 240–42). More significant is the semi-Isma'ili tenor of the *Epistles* of the Old Brethren themselves; yet they were not lettrists. By contrast, Isma'ili influence is likewise detectable in both Ibn al-'Arabi's works and those of the Ḥurūfiyya (Ebstein, *Mysticism*; Mir-Kasimov, *Words of Power*, pp. 166 n6, 414–16). But while the former was embraced as foundational authority on lettrism by the New Brethren, they execrated the latter group as perversers of that science and a political thorn in their side. At the same time, contemporary pro-occultist Sunni authors like the famed historian Taqī l-Dīn al-Maqrīzī (d. 1442) – Ibn Khaldūn's sole student – could go so far as to hold up Ḥasan-i Ṣabbāḥ (d. 1124) himself, dread Lord of Alamut, as exemplar of Islamic occult-scientific kingship, thereby flying in the face of his mentor's anti-occultist and anti-Isma'ili project (Melvin-Koushki, "In Defense of Geomancy," p. 370). As that may be, any insistence that the New Brethren were simply reprising Isma'ili precedent can only be polemical and antihistorical: these outspokenly Sunni Pythagorean imamophiles are the first movement in Western history as a whole to champion, develop and propagate lettrism (with astrology) as the master framework for all human knowing and doing, Islamic or otherwise.

He failed, of course, and his rivals succeeded beyond his wildest fears; Ibn Khaldūn's painfully simplistic dualist-cyclicist theory of world history – meant to subvert similarly cyclicist but less dualist astrological theory – was panned by contemporaries and ignored for centuries as terminally unreliable, antihistorical and puritanical, badly out of step with the most exciting intellectual developments of his day. The fact that the Tunisian historian was only finally seized upon and feted by French colonialist scholars in the nineteenth century, whence his bizarrely disproportionate fame in modern scholarship and the total disappearing of the far more influential New Brethren of Purity, is one of the many great ironies of contemporary Orientalism.²⁹ More modern ink has been spilled on this single anti-occultist polemicist than all of the occult sciences combined. Yet the latter formally constitute at least *half* of the natural and mathematical sciences in the Islamicate archival record, and hence over ten percent of the many millions of surviving manuscripts and artifacts, despite the combined depredations of colonialism, reformism, nationalism and salafism over the last two centuries.³⁰ When we understand the true import of their standard Persianate designation as *'ulūm-i gharība* or the *weirding sciences* par excellence – weirdness (*gharābat*) being a category central to aesthetics and epistemology itself from Naṣīr al-Dīn Ṭūsī onward, and predicated on the concept of *wonder* (*ta'ajjub*) as literary, religious, ethical, philosophical and scientific imperative – that percentage massively increases to encompass many other genres besides.³¹ To translate such perverse disproportionality into current historiography of science terms, this is to scorn physics in favor of phrenology – and then to pretend neither has any cultural or political relevance. Five centuries after his death, Ibn Khaldūn has in the end succeeded in his life's project through alliance with the colonizers: to write post-Mongol Islamic (occult) philosophy and (occult) science out of Western intellectual history, to burn Muslim witches and their books.

To restore them to the same, to rescue them from Bruno's Inquisitorial pyre, we need only turn to the New Brethren, who embraced such smears as a point of pride. Indeed, it was by way of polemical retort that they styled themselves such, thereby broadcasting their intent to revive and expand the occult-philosophical project of the Old Brethren of 'Abbasid Iraq – that anonymous

29 Ibn Khaldūn's theory of dynastic cycling did enjoy a modest reception in later Ottoman historiography as well, but only as support to similar models independently developed for the Ottoman case during the sixteenth century; see Fleischer, "Royal Authority"; and below.

30 Melvin-Koushki, "Is (Islamic) Occult Science Science?," pp. 307, 311.

31 Melvin-Koushki, "Naṣīr al-Dīn Ṭūsī"; idem, "Translating Esotericism"; idem, "Definition as (De)colonial Weapon"; Zadeh, *Wonders and Rarities*.

semi-Ismaʿīlī and wholly Pythagorean group of encyclopedist-occultist bureaucrats that appears to have claimed Maslama al-Qurṭubī (d. 964) himself, possible author of the watershed Andalusian Arabic grimoire *Goal of the Sage* (*Ghāyat al-ḥakīm*, Lat. *Picatrix*), as a member.³² Also unlike the Old Brethren, however, but like Ibn al-ʿArabī and Aḥmad al-Būnī, their early fifteenth-century Sunni heirs seized upon *lettrism* as the primary vehicle for a new brand of Pythagorean-Platonic occult philosophy-science encompassing and transcending Peripateticism, Illuminationism and Sufism as discrete socio-intellectual currents. Emphatically perennialist and universalist, and far more explicitly imperialist, this philosophy asserted *letter-number* to be the supreme object of metaphysics and physics in equal measure, the *coincidentia oppositorum* marrying Plato and Aristotle by means of a Semiticized and ʿAlidized Pythagoras.³³ This marriage of all fields and means of human knowing is encapsulated in their more usual self-designation as *ahl-i taḥqīq*: experimentalist investigators into reality as such.³⁴ A virtually identical development, as noted, defined the occult philosophy of that other Renaissance, particularly that of Cusanus, Pico and Dee – the would-be Ibn Turkas of Latin Christendom.

A New Metaphysics – and Living Philosophy as Magic

This Arte [scil. Archemastrie = kabbalah], teacheth to bryng to actual experience sensible, all worthy conclusions by all the Artes Mathematicall purposed, & by true Naturall Philosophie concluded ... *Science* I may call it, rather, then an Arte: for the excellency and Mastershyp it hath, over so many, and so mighty Artes and Sciences. And bycause it procedeth by *Experiences*, and searcheth forth the causes of Conclusions, by *Experiences*: and also putteth the Conclusions them selves, in *Experience*,

32 See Fleischer, “Ancient Wisdom”; Binbaş, *Intellectual Networks*, pp. 106–13; de Callatay, “Magia en al-Andalus.” Since those medieval Arabic texts randomly and often poorly done into Latin continue, predictably, to be taken as index of all Islamic Magic, it bears emphasizing that the *Ghāya* never acquired the same cachet in its original Arabic (much less in Persian) as it did in Latin, where due to general bibliographical impoverishment it loomed much larger.

33 A move in the imperial-occultist direction so roundly embraced by the New Brethren may be seen, however, in the long version of the Old Brethren’s concluding epistle on magic, which advocates for the use of occult science as *sine qua non* for sacral kingship.

34 Binbaş, *Intellectual Networks*, pp. 96–104. For Ibn al-ʿArabī in particular *ʿilm al-ḥurūf* and *taḥqīq* are synonyms.

it is named of some, *Scientia Experimentalis*. The *Experimentall Science*.
Nicolaus Cusanus termeth it so....

JOHN DEE, *Mathematicall Præface*

Our Timurid “Pythagoricall, and Platonick perfect scholer”³⁵ is the most prominent member of the venerable Turka family of Isfahani scholars and judges, from whom he inherited the position of Shafi’i chief judge of Isfahan and Yazd. Their name, meaning “Little Turk,” commemorates the family’s origins in Khwarazmshahid-Seljuq Khujand, in what is now Tajikistan. His grandfather, Šadr al-Dīn Abū Ḥāmid Muḥammad Turka (fl. early 14th c.), was a noted philosopher of the Ilkhanid period, and among the first to attempt to synthesize Peripatetic and Illuminationist philosophy with the radical monism of Ibn al-ʿArabī – a project his grandson pursued even more influentially. This synthesizing ethos, of course, defines post-Mongol Islamic philosophy generally, and especially its culmination in Safavid Iran. As a major systematizer and promoter of Akbarian metaphysics, modern scholarship has typically styled Ibn Turka a key transitional thinker linking Naṣīr al-Dīn Ṭūsī (d. 1274) and Mullā Šadrā (d. 1636). It singles out his *Clarifying the Principles* (*Tamhīd al-Qawāʿid*), a commentary on his grandfather’s *Principles of Divine Unity* (*Qawāʿid al-tawḥīd*), treating of absolute existence (*al-wujūd al-muṭlaq*), as a masterpiece of Avicennan-Suhrawardian-Akbarian synthesis, one that remains an integral part of Twelver Shi’i seminary curricula even today. On the strength of this synthesis, the renowned Shi’i philosopher-theologian ʿAllāma Ṭabāṭabāʾī (d. 1981) ranked Ibn Turka’s importance within Islamic intellectual history as equal to that of the great Neoplatonizers Fārābī (d. after 942) and Suhrawardī (d. 1191).³⁶

Yet Doctor Little Turk’s stated project was not simply to synthesize, but to theurgically transcend. In true early modern millenarian fashion, he sought to create a new lettrist metaphysics-physics that would render Avicennan and Suhrawardian philosophy obsolete: letter-number as *coincidentia oppositorum* (*majmaʿ al-aḍḍād*) and hence sole basis for a universal, monist science (*ʿilm al-tawḥīd*), the intellect’s only vehicle of return to the One.³⁷ (That Nicholas of Cusa, Ibn Turka’s later contemporary, coined the Latin term suggests it as a

35 Dee, *Mathematicall Præface*, verso of first page.

36 For a full biography and study of Ibn Turka, see Melvin-Koushki, “The Quest.”

37 I have elsewhere shown the *coincidentia oppositorum* to be the central theme of Islamic occult philosophy in general and lettrism in particular from the early fifteenth century onward; see Melvin-Koushki, “Of Islamic Grammatology.”

major node for comparative study.³⁸) Epitomizing this doctrine, the quranic dictum *He is the First and the Last, the Manifest and the Occult* (Q 57:3) runs throughout his oeuvre and those of the New Brethren for centuries after as watchword and motto.

Specifically, Ibn Turka flatly rejects both the Peripatetics' Existence (*wujūd*) and the Illuminationists' Essence (*māhīyya*) as illegitimate – because terminally dualist – categories of metaphysical analysis, replacing them with the mathematical Letter (*ḥarf*), identical to the Illuminationists' Light (*nūr*). The cosmos is thus constellated not by *tashkīk al-wujūd*, the transcendental modulation of Being – a concept first cast in *semantic* terms and fundamentally *weirded* by Naṣīr al-Dīn Tūsī in his seminal commentary on Ibn Sīnā's (d. 1037) *Pointers and Reminders* (*al-Ishārāt wa-l-tanbīhāt*) –, but by the transcendental modulation of the Letter as uncreated, all-creative Light (*tashkīk al-ḥarf*), gradually emanating down from the One in extramental (*'aynī*), mental-mathematical (*lubābī iḥṣā'ī*), spoken-aural (*kalāmī*) and finally written-visual (*kitābī*) form.³⁹ It is in view of his revolutionary challenge to and subversion of mainstream Arabic philosophy that Muḥammad-Taqī Dānishpazhūh (d. 1996) hailed Ibn Turka the "Spinoza of Iran."⁴⁰ But where Spinoza's sensory theory is neutral and pragmatic, and hardly mathematizing, Ibn Turka's upends the Avicennan-Aristotelian schema, making Sight superior to Hearing as highway back to celestial, mathematical Mind – hence observational astronomy (among other sciences) as theurgic imperative.⁴¹

This radical project culminated in his *Investigations*, the first summa, as noted, of Islamic Pythagoreanism and centerpiece of Ibn Turka's oeuvre. Therein he pointedly revives and updates the Pythagorean, occult-scientific platform of the Old Brethren of Purity by recasting it in purely lettrist, dual-monist, Akbarian-Ḥamūyan terms. Our Isfahani occult philosopher states that his primary purpose in this work is to demonstrate the roots of all manifestation in the One and schematize the derivation of the Many therefrom. This information, in turn, will allow the adept to manipulate letter-number – the uncreated, all-creative matrix through which the One self-manifests – to access and control every epistemological and ontological level of the cosmos, thus constituting a metaphysics-physics continuum from rarefied letter theory

38 See especially Albertson, *Mathematical Theologies*. Cusanus has lately been received by modern European philosophers and theologians as a thinker to think with (Albertson, ed., *Cusanus Today*); why not show the far more influential Ibn Turka the same intellectual hospitality?

39 Melvin-Koushki, "Of Islamic Grammatology," pp. 60–62.

40 Dānishpazhūh, "Majmū'a-yi rasā'il-i Khujandī," p. 312.

41 Melvin-Koushki, "Pythagorean Elevation."

to practical computing and other talismanic technologies. The supreme dignity of its object necessarily renders lettrism the supreme science:

The subject of the science we have here in view is the One (*al-wāḥid*) insofar as it is one, regardless of the form in which it manifests in all the variety of its significations. The all-pervasive, all-encompassing nature of One with respect to existence being obvious, this science is therefore necessarily superior to all other sciences by an order of magnitude.⁴²

It enjoys this status, moreover, precisely because it is the most *mathematical* of all the sciences:

Behind this gate [i.e., the new lettrist theory here proposed] are the treasuries of the Seal [Muḥammad's] glory and the protected space of his intimacy which contain necklaces of precious jewels (*ʿuqūd farāʿid al-jawāhir*) and all else laid there in store. [Muḥammad] collected all this and provisioned therewith his son [ʿAlī b. Abī Ṭālib], the Seal of sacral power (*walāya*) and standard-bearer of understanding and guidance. These necklaces (*ʿuqūd*) are numerical knottings (*al-ʿuqūd al-ʿadadiyya*), the spiritual-intellectual form of the Book that was sent down from the highest Pen to the holy Tablet. Number (*ʿadad*), then, is the best means of acquiring sciences of great benefit and numerous as grains of sand, the primordial mine preserving the gems [at the core] of all the standard and mainstream sciences ... Let the researcher therefore set aside his various misconceptions and inquire into the matter of number, for it is the fountainhead of all the sciences, the quarry of all realities, an ocean of insights both manifest and occult.⁴³

In short, Ibn Turka here completes the Islamicization of occult-scientific Pythagoreanism, a process begun by the Old Brethren – their *Epistles* likewise founded on the premise that “the science of number is the root of the sciences, the essence of wisdom, the foundation of knowledge and the [principal] element of all things”⁴⁴ –, by associating it definitively with ʿAlī (d. 661), the first

42 *Kitāb al-Mafāḥiṣ*, Tehran, MS Majlis 10196, f. 53b; see Melvin-Koushki, “The Quest,” p. 331.

43 *Kitāb al-Mafāḥiṣ*, Tehran, MS Majlis 10196, ff. 52a, 56b; Melvin-Koushki, “The Quest,” pp. 338, 344. Note that *ʿadad*, translated here as ‘number,’ is also the standard term for arithmetic as part of the quadrivium.

44 *Rasāʾil*, 1/21–22; trans. Endress, “Mathematics and Philosophy,” p. 133.

Shi'i Imam and Selenic shaman-king fountainhead of occult science for the Islamic dispensation.

But our staunchly, outspokenly Sunni occult philosopher must not be facilely pigeonholed as a Shi'i on this basis (for all that he routinely has been to date).⁴⁵ Like his later Christian kabbalist-Pythagorean peers, Ibn Turka was rather a committed perennialist, explicitly invoking lettrism – precisely as Pico would kabbalah – as core and culmination of the *prisca theologia*, which he associates in the first place with Solomon, Pythagoras and Plato. Unlike them, he was also a progressivist: our Sunni-Shi'i occult philosopher draws an ontological equivalency between the Shi'i Imams and the ancient Greeks in general and 'Ali and Pythagoras in particular, and further suggests, astonishingly, that moderns too may and should aspire to the same status.⁴⁶ His *Investigations* thus functions not as proof of the author's Shi'ism, but *manual for becoming a Pythagorean transhuman* – hardly the orthodox Twelver position!

It is all the more striking, therefore, that Ibn Turka's perennialist-progressivist, Pythagorean-Imamic, openly transhumanizing lettrism was most eagerly embraced precisely by the committed Twelver philosophers of the Safavid period – as a primary *Sunni* means of *Shi'izing Iran*.⁴⁷ Chief among them is Mīr Dāmād, imperial Safavid philosopher extraordinaire and Chief Jurisconsult, who died in 1630, the same year as his fellow Pythagoreanizer Kepler. Explicitly embracing the Ibn Turkian legacy, and particularly the *tashkīk al-ḥarf* schema, Mīr Dāmād wrote three works wholly or substantially lettrist in approach. The most important of these is *Firebrands and Epiphanies* (*Jazavāt u mavāqīt*), his only major work in Persian, composed at the instance of his patron Shah 'Abbās the Great (r. 1587–1629), which offers a full, and characteristically ornate, treatment of his philosophical system within a Pythagorean-lettrist framework, centered on his signal concept of *metatime*.⁴⁸

Here as elsewhere, Mīr Dāmād – hailed by his contemporaries the Third Teacher (*mu'allim-i sālis*), after Aristotle and Fārābī – compares and contrasts what he calls his Yemeni or oriental philosophy (*al-ḥikma al-yamāniyya*) with Greek precedent (*al-ḥikma al-yūnāniyya*).⁴⁹ The former is his trademark Avicennan-Illuminationist-Akbarian-Twelver synthesis; the

45 Melvin-Koushki, "The Quest," pp. 69–77.

46 Ibid., ch. 6.

47 Melvin-Koushki, "Occult Ecumenism."

48 For a discussion and contextualization of these works, see Melvin-Koushki, "World as (Arabic) Text"; Terrier, "The Wisdom of God." On the Safavid reception of Pythagoras generally see idem, "La figure de Pythagore."

49 Esots, *Patterns of Wisdom*, ch. 1; Rizvi, "Mīr Dāmād in India," p. 15: "Often he presents his argument by stating that he will first examine the 'Greek' philosophical position and then move on to the Yemeni one."

latter, significantly, is here represented in the first place by Pythagoras and the Pythagoreans, called by him *fīṣāghūrasīyyīn* or *āl-i Fīṣāghūras*, together with Ps.-Aristotle (i.e., Plotinus) and Plato.⁵⁰ (Like Ibn Turka and other lettrists, Mīr Dāmād takes Pythagoras to be Solomon's contemporary and disciple.⁵¹) The camp of the Greek ancients is rounded out with citations of Nicomachus, Archytas, Anaximenes, Ptolemy, Homer and the Stoics. In the opening section of the *Jazavāt* Mīr Dāmād further declares the pseudepigraphal *Theology of Aristotle* (*Uthūlūjīyā*, Lat. *Theologia Aristotelis*) – the immensely influential tenth-century Arabic paraphrase of Plotinus's *Enneads* IV–VI – to be fundamental to his entire philosophical system.⁵² It is only in the Yemeni or Islamic dispensation, however, that metaphysical perfection may be attained, that the potentialities latent in Greek precedent may be fully actualized.

Yet that perfection and actualization remains forever *Pythagorean*. In the *Firebrands* Mīr Dāmād offers this (uncharacteristically clear) summary of his cosmology as a whole – perhaps the concisest epitome of lettrist Pythagoreanism ever penned in the Greater Western tradition:

In sum: The realm of the letter corresponds to that of number, and the realm of number corresponds to that of engendered existence (*kawn*); and the relationships (*munāsabāt*) that obtain in the realm of letter correspond to those that obtain in the realm of number, which in turn correspond to the commixtures (*mumāzajāt*, *mukhālaṭāt*) that constitute the realm of engendered existence.⁵³

Like his equally committed Christian kabbalist colleagues just to the west, moreover, the Third Teacher was very much alive to the scientific implications of his thoroughgoing Pythagoreanism. For the goal of Pythagorean-Platonic philosophy is *theosis* (*ta'alluh*), self-divinization; and theosis is no armchair

50 See e.g. *Jazavāt u mavāqīt*, pp. 46, 47, 60, 62–5, 79, 105, 110, 115, 119–20, 134, 143, 147, 177, 179, 186, etc. Pythagoras and Plato in particular are routinely identified with theosis (*ilāhī*, *muta'allih*). Mīr Dāmād's primary source for Pythagorean doctrine is the famed heresiographer Shahrastānī's (d. 1153) description thereof in the *Book of Religions and Creeds* (*Kitāb al-Milal wa-l-niḥal*), quoting from it directly on pp. 177–79; on p. 135, moreover, he counters Shahrastānī's objection to the Pythagorean concept of the correspondence of letters to reality.

51 *Jazavāt*, p. 46; cf. Melvin-Koushki, "The Quest," pp. 318, 325. Cf. Bruno's similar association of Pythagoras with Solomon, and Pico's assertion that Pythagoras was a Jew – hence the inextricability of his doctrine from kabbalah (Joost-Gaugier, *Pythagoras*, pp. 42, 55).

52 *Jazavāt*, p. 15.

53 *Ibid.*, pp. 133–34.

pursuit, but a strenuous, traumatic, experimental, *scientific* praxis – one frequently ritual-magical and theurgical in nature. While this basic point has often been lost on modern historians, whether positivist or religionist, it cannot be overemphasized that Islamic philosophy, for Ibn Turka, Mīr Dāmād and many other major early modern thinkers, was a total way of life;⁵⁴ and its primary mode of praxis – preferably out-of-body – was precisely Pythagorean-Imamic *occult science*.⁵⁵

Needless to say, such an occultist conception of philosophy has very deep roots in the Western tradition, and may well go back to the very “father of philosophy,” Pythagoras, himself. But the multiple Renaissances of Platonism and Pythagoreanism that so define Western early modernity naturally gave rise to a new golden age of Weirding Science unprecedented in technological and imperial scope – and especially in its Islamic subset.

Now Islamic philosophy had been heavily Neoplatonic since its inception, as is well known, particularly due to the fusion of Aristotle with Plotinus and Proclus in their Arabic receptions. The unparalleled importance of Ibn Sīnā (Avicenna, d. 1037) – that second Aristotle – to all subsequent Western philosophy is due precisely to his definitive synthesis of Neoplatonism and Aristotelianism, which synthesis became hegemonic under his name thereafter. But the Iranian sage was no booster of the occult sciences (he rejects astrology and alchemy in particular), and was certainly no Pythagorean. By contrast, his challengers Suhrawardī, father of Illuminationism, and Fakhr al-Dīn Rāzī (d. 1210), amalgamator of Avicennism, Ash‘arism and occult science, reasserted astral-magical ritual and planetary ascent as the philosopher’s primary business.⁵⁶ Among the most telling indexes of the intellectual-cultural seachange wrought by Ibn Turka and his heirs, whereby philosophy was again conceptualized and lived as a theurgical-magical practice, but now in lettrist terms, is therefore Mīr Dāmād’s transmogrification of Ibn Sīnā himself into preeminent Pythagorean lettrist *magus*, and even the pseudepigraphal author of *Spellcasters’ Treasures* (*Kunūz al-mu‘azzimīn*), one of the most popular Persian grimoires of the Safavid era. He likewise was the first to ardently promote the (possibly authentic) *New Year Treatise* (*Risāla Nayrūziyya*), a concise Avicennan-*cum*-lettrist answer to late antique Neopythagorean cosmology.⁵⁷

54 Rizvi, “Philosophy as a Way of Life.”

55 Melvin-Koushki, “Mīr Dāmād’s *On Doffing*.” While less explicit as to the nature of his occult-scientific practice, Suhrawardī, as a good Neoplatonist and perennialist, is here an important precedent. See e.g. Walbridge, *The Leaven of the Ancients*, ch.5; on the place of Pythagoras in Illuminationist philosophy see *ibid.*, ch. 6.

56 Piątak, “Philosophy, Mysticism and Magic”; Noble, *Philosophising the Occult*.

57 Melvin-Koushki, “World as (Arabic) Text.”

Avicennism and lettrism would remain married under the Pythagorean banner for centuries after; the greatest metaphysicians of the early modern Persian cosmopolis were also its greatest mages.

An equally telling index of the same seachange is the revival of popularity of the *Epistles* of the Old Brethren of Purity themselves after centuries of neglect, particularly in their abridged Persian translation, titled simply *Epitome of Philosophy* (*Mujmal al-ḥikma*). This revival is explicitly testified to by Mīr Dāmād's Shirazi colleague Abū l-Qāsim Anṣārī Kāzīrūnī (fl. 1605) – a student, not incidentally, of Ibn Turka's great-great-grandson Afzal al-Dīn b. Ḥabīb Allāh Turka (d. 1583) – in the latter's remarkable *Ladder to the Heavens* (*Sullam al-samāwāt*).⁵⁸ An eclectic, genre-bending work equal parts theological tract, history of philosophy, literary history and encyclopedia of the sciences, the *Ladder* served Safavid imperial needs admirably by constructing an intellectual-religious pedigree for Shah 'Abbās that is simultaneously Twelver Shi'i, Sufi, occult-scientific, Pythagorean and Platonic: an impressive feat of synthesis indeed, and one reflective of the shah's experimentalist efforts to create a new, more stable Safavid imperial culture. And the *Epistles*, it would seem, were a crucial component of that culture. Of them Kāzīrūnī declares:

There is not a single word in the *Epistles* [of the Brethren of Purity] that this wretch has not read repeatedly with the eye of desire and longing ... In some of these epistles it is stated that on most topics their discussion accords with the doctrine of Pythagoras.⁵⁹

While Ibn Turka and his fellow New Brethren of the early fifteenth century were not engaged in textually excavating Greek philosophy in the way that became standard in the later Safavid period, then, and while they were most certainly not Shi'i, the philhellenizing lettrist revolution they spearheaded helped shape the intellectual and imperial terrain of Mamluk Egypt, Timurid Iran and Ottoman Anatolia and continued to reverberate across the broader Persianate world for centuries thereafter. Thus was the Old Brethren's Pythagoreanizing, Weirding project restored to the Western philosophical and cultural mainstream – and there it remained to the era we are pleased to call modern.

58 Melvin-Koushki, "The Quest," pp. 43–45; idem, "Powers of One," pp. 164–66.

59 *Sullam al-samāwāt*, p. 205. Kāzīrūnī believes the *Epistles* to be the work of a single author who died sometime between 500–50 AH (1107–55 CE) – i.e., that they are a firmly twelfth-century work –, and implies that he consulted rather the *Mujmal al-ḥikma* (pp. 205, 207).

A New Physics: Mathematizing the Cosmos

Raising towards heaven our cabbalistic eyes....

JOHN DEE, *Monas Hieroglyphica*⁶⁰

The cosmological doctrine of the Two Books, scripture and nature, has been feted by historians of science as a guiding theme for the emergence of “scientific modernity” – the upshot of western Europeans (and no one else) reading the world as mathematical-linguistic text. The kabbalistic decoding of this text becomes *science*; its recoding, originally by way of magic, becomes *technology*.

Yet earlier and contemporary Muslim Pythagorean occultists were no less committed to reading the world as mathematical-linguistic text, including in the first place Ibn Turka and his friends and heirs.⁶¹ Ibn Turkian lettrism, quite simply, runs the epistemological gamut from mathematics to magic, metaphysics to physics, philosophy to science; it demands as actualization of the Hellenic-Abrahamic legacy the mathematization of an essentially textual and hence conscious cosmos. To what extent, then, should we now incorporate Muslim occult philosophers into our *mathesis* narrative of the scientific rise of the West?

To at least the same extent as their infinitely better studied Christian cognates have been. For Ibn Turka is the first in the Islamic tradition, and the Western more broadly, to successfully install lettrism as the most immediate intellectual and political context for the pursuit of astronomy, queen of the quadrivium. At his urging, his royal patrons and friends – including several of the greatest astronomers and philosopher-kings of the fifteenth century – fully mathematized that discipline. That is to say, they finally made so bold as to dispense with the hampers of Aristotelian physics, which requires a stationary earth, and hence grounds the geocentric model: the freeing of astronomy from philosophy, as Jamil Ragep has summarized their project.⁶²

As historians of science have now shown, this astronomical seachange – the most immediate and obvious precedent for the Copernican revolution to the west – was accomplished by Ulugh Beg b. Shāhrukh (r. 1409–49) and the coterie of mathematician-astronomers the Timurid ruler installed at his famed observatory and madrasa complex in Samarkand, including in the first place

60 Trans. Josten, p. 175. The original Latin: “Sublevatis ergo in Caelum oculis Cabalisticis....”

61 Arabic, it should be noted, lent itself rather more readily to this “nature-scriptural consciousness” than Greek and Hebrew, neither of whose alphabets have enough letters to fully represent units, tens and hundreds, up to one thousand.

62 Ragep, “Freeing Astronomy.”

Qāzizāda Rūmī (d. 1432) and his student ‘Alī Qūshchī (d. 1474). The latter, a second Aristarchus (d. 230 BCE), pushed a radically mathematical model that allowed for a rotating earth. (His contemporary and fellow Pythagorean Nicholas Cusanus (d. 1464) did the same.⁶³) And together they oversaw the production of Ulugh Beg’s *New Sultanic Star Tables* (*Zīj-i jadid-i sulṭānī*), the most accurate astronomical (and astrological) pre-telescopic handbook ever produced and model from London to the Strait of Malacca down to the nineteenth century.⁶⁴

But what motivated this generation of Muslim star scientists to become the first in recorded history to achieve arcminute precision? Why the unprecedented need for mathematical accuracy? The current historiography of science in incurious, preferring instead to abstract these Timurid astronomers from their lived, sociopolitical context – a context in which lettrists and mathematician-astronomers (sometimes the same person) publicly professed a common, expressly Pythagorean purpose, maintaining a brotherly (*ikhwānī*) correspondence with one another and sharing their treatises to this end. Ibn Turka and his student and closest friend, Sharaf al-Dīn ‘Alī Yazdī (d. 1454), the Timurid dynastic historian, litterateur, lettrist-astrologer and mathematician, were friends and colleagues to Qāzizāda Rūmī; Yazdī even worked at the observatory for a time.⁶⁵ In such a context, in other words, it was only natural for a Pythagorean like Qāzizāda – or Kepler after him – to seek to mathematize the cosmos; and his warm friendship, from childhood, with Ibn Turka, and association with Akhlāṭī himself, cannot but have shaped his thinking.⁶⁶ Indeed, Ibn Turka states this fact quite explicitly, and claims the Samarkand Observatory as a realization of the New Brethren’s project. More, Ibn Turka fashions Ulugh Beg ultimate philosopher-king and sultan-scientist (*al-sulṭān al-faylasūf*), the Archytas of the age, for whose advent he had long been waiting. (This he did in *circa* 1426 in the dedicatory preface of a lengthy, philosophically dense Arabic lettrist work, *Sharḥ al-basmala*, as yet unedited and unstudied; Qāzizāda

63 Joost-Gaugier, *Pythagoras*, p. 134.

64 See e.g. Ragep, “Copernicus and His Islamic Predecessors”; Saliba, *Islamic Science*; Fazlhoğlu, “Between Reality and Mentality.”

65 Binbaş, *Intellectual Networks*, p. 65. That this social and scholarly connection continued to be well-known for at least the next century is further suggested by a book painting in a deluxe early Safavid copy of a late Timurid biographical dictionary, Kamāl al-Dīn Ḥusayn Gāzurgāhī’s (fl. 1503) *Lovers’ Seances* (*Majālis al-‘ushshāq*), which depicts Ibn Turka, Qāzizāda and Yazdī as a seated triumvirate together consulting a book, in accompaniment to the entry on Akhlāṭī; see Paris, BnF, Ms Supplément persan 775, f. 116a.

66 Melvin-Koushki, “Powers of One,” p. 136 n. 32. On Kepler as committed Pythagorean see e.g. Hallyn, *The Poetic Structure of the World*.

Rūmī's effusive letter thanking him for a copy is extant.⁶⁷) Similarly, it is here of signal importance that our Isfahani occult philosopher began pushing precisely for a return to a *mathematical* cosmology, this in his *Investigations*, in 1420 – the same year construction of the observatory was begun.⁶⁸

That this Timurid crew did not themselves explicitly champion the classically Pythagorean doctrine of *Heliocentrism*, as their later Latin Christian counterparts would gradually do over the following centuries, has everything to do with their reduction to a mere footnote (at best) in the historiography of science. Yet the Copernican model itself was conceived of and embraced in specifically Pythagorean, occult-scientific and Christian terms by its defenders and foes alike; thus a critic of the stature of Athanasius Kircher (d. 1680), the great Jesuit polymath-occultist, could mock Kepler's geometrical version thereof as "a Copernico-Pythagorean contrivance."⁶⁹ An inveterate Pythagorean contriver too, Ibn Turka was rather a *Selenocentrist*: 'Alī as Pythagoras is Letter as Light as Moon. But it was precisely the 'Alid Selenocentrism of Timurid and then Safavid imperialism (Temūr as Moon conquerer to Alexander's Sun) that made Mughal and Ottoman imperialism so thoroughly Jesuan and Heliocentrist in response – and to a far greater degree than any contemporary Latin Christian polity. Early modern Persianate cultures were thus predicated on Selenocentrism and Heliocentrism, whose lettrist, binary play is the soul of Pythagoreanism; only the colonialist-Orientalist coding of Islam as strictly Religion has erased this simple fact from the story of Science.⁷⁰

The New Brethren's Pythagoreanizing-mathematizing project is medicine for this historiographical ill: it must now be approached as of a conceptual and sociopolitical piece with that of both the Samarkand astronomers and the Copernicans in western Europe (the likely lack of direct textual transmission to the latter notwithstanding); the Western cultural-genetic continuities are simply too obvious to be missed or explained away. We now know that the holy lineage *Copernicus-Kepler-Galileo-Newton* did not simply suddenly spring forth from the arid scholastic ground of Latin Christendom, a virgin birth, but was an organic outgrowth of seven centuries of incremental developments in Arabic astronomy-astrology. But its natural culmination in Newton's *Principia Mathematica* (1687), that cornerstone of scientific modernity,

67 On this work and its preface see Melvin-Koushki, "The Quest," pp. 111–12; for Qāzizāda's letter see *ibid.*, pp. 146–47, 569–70. On the term *al-sultān al-faylasūf* see Fazlıoğlu, "The Samarqand Mathematical-Astronomical School," p. 46.

68 Melvin-Koushki, "Timurid-Mughal Philosopher-Kings."

69 Rowland, "Athanasius Kircher," 192. See Westman, *The Copernican Question*; Joost-Gaugier, *Pythagoras*.

70 Melvin-Koushki, "Selenocentrism and Heliocentrism."

whereby physics too was fully mathematized, only becomes historiographically explicable within this Greater Western Pythagorean-lettrist framework. That Euro-American intellectual historians continue to focus monomaniacally on the former but elide the latter as Oriental, medieval and mystical irrelevance is merely an artifact of our own rupturous vogue as self-styled moderns. The term “Scientific Revolution” itself is of strictly twentieth-century vintage, and reflects rather the ruptures of that apocalyptic, millenarian, totalitarian, Neomanichean age.

But the project of our early modern Pythagoreans, Muslim and Christian, was no less millenarian and universalist. As a case in point: Ibn Turka's radically new hierarchicalization of the religious and rational sciences – in ascending order: traditionism, theology, Peripateticism, Illuminationism, Sufism, lettrism – epitomizes his millenarian ethos more generally, simultaneously perennialist and progressivist-supersessionist, and above all theurgic. According to Ibn Turka's Selenocentric reclassification thereof, moreover, lettrism, as a Pythagorean-Platonic mode of knowledge intrinsic to the prophetic-philosophical legacy throughout history and operative on all levels of reality, is alone capable of serving as a universal science organizing all of human knowledge under its aegis.⁷¹ To borrow Vittoria Perrone Compagni's description of the not dissimilar objective of Heinrich Cornelius Agrippa (d. 1535) in his similarly landmark *Three Books of Occult Philosophy* (*De occulta philosophia libri tres*), Ibn Turka “proposes a total reorganization of [lettrism] as an umbrella science which, by gathering under a single roof all the cognitive data collected in the various fields of scientific research, would guarantee the effectiveness of each branch of research and make explicit its potential for acting upon reality.”⁷²

Triumphalist teleologies notwithstanding, then, it is remarkable that, in the absence of direct contact, the quest for a universal science was universally pursued along Pythagorean, kabbalist-lettrist lines throughout the Islamo-Judeo-Christian world during the early modern period – a trend that became mainstream significantly earlier in the Persianate context, where the cosmos was first mathematized.⁷³

A second occult science likewise stands as index of this epochal renaissance of Pythagoreanism throughout the early modern Greater West: geomancy, ‘sand science’ (*ilm al-raml*). This divinatory art, Arabic cognate to the *I Ching* (both are based on a binary even-odd system), was third in popularity only to astrology and oneiromancy across the Afro-Eurasian ecumene, and is still

71 Melvin-Koushki, “The Quest,” ch. 6.

72 See the introduction to her edition of *De occulta philosophia libri tres*, p. 16.

73 Melvin-Koushki, “Powers of One.”

widely practiced today from Iran and Yemen to West Africa and Madagascar. It would even be seem to be making a serious comeback globally among anglophones.⁷⁴ Of Afro-Islamic origin, and first consolidated in Arabic in the tenth century and done (poorly) into Latin in the twelfth, the science reached its greatest degree of complexity and sophistication in the post-Mongol Persianate imperial context. There it was widely lionized as a technological means of controlling history and society and one's political and military fate; geomancers accordingly emerged as influential fixtures at Mughal, Safavid and Ottoman courts. Some rulers, such as the Safavid shah Ṭahmāsb I (r. 1524–76), even became ardent practioners in their own right. As bestselling manuals of the era make clear, geomancy was in effect an early modern form of googling, social networking, wiretapping, trade and weather forecasting, job finding, matchmaking, CT scanning, ultrasounding, metal and lie detecting, home surveillance and radar.⁷⁵

As such, geomancy exercised some of the best minds of the early modern Persianate world as a mainstream *mathematical* science – one constituting scientific, empirical proof for the validity of Pythagorean-Platonic panpsychist cosmology.⁷⁶ Predictably, these included the New Brethren and their associates in Samarkand. Akhlāṭī himself appears to have authored a popular Persian geomantic manual, *The Treatise of Surkhāb*; one of his students, Ḥasan Abarqūhī (fl. early 15th c.), achieved fame solely as a geomancer; Yazdī wrote a tract in defense of the same science, arguing for its validity with lettrist and other proofs; Ulugh Beg is reported to have regularly consulted its practitioners; and 'Abd al-Qādir Rūyānī Lāhījī (d. 1519), student of 'Alī Qūshchī and author of several astronomical texts, likewise penned an important Persian geomantic manual, *Key to The Keys (Miṭāḥ-i Mafāṭih)*.⁷⁷

It was one of their Safavid successors, however, who first argued for the science's *indispensability* to the modern Pythagorean-Platonic cosmologist: Shams al-Dīn Khafī (d. 1535). Peer and contemporary to Kepler – and father, not incidentally, of Maḥmūd Dihdār Shīrāzī (fl. 1576), the most prolific Persian

74 See e.g. Greer, *The Art and Practice of Geomancy*; Melvin-Koushki and Cummins, "Geomancing is Googling."

75 Fleischer, "Shadow of Shadows," 60; for more popular Ottoman usages see Melvin-Koushki, "Divining Past, Present and Future."

76 The eminent Ash'ari theologian, philosopher and occultist Fakhr al-Dīn Rāzī (d. 1209) was the first Arabo-Persian encyclopedist to reclassify it as such; see Melvin-Koushki, "Powers of One," pp. 145–47; Noble, *Philosophising the Occult*.

77 Melvin-Koushki, "Persianate Geomancy"; see also Şen, "Astrology in the Service of the Empire," p. 177 n. 29. Yazdī's tract in translated in Melvin-Koushki, "In Defense of Geomancy."

author on lettrism of the sixteenth century, and self-identified successor to Akhlāfī –,⁷⁸ Khafī has been hailed by historians of science the foremost Muslim planetary theorist of the same century.⁷⁹ What has not been recognized in modern scholarship is the fact that in one of his two treatises on geomancy, titled simply *Risāla dar raml*, he makes this point with unprecedented explicitness: it is the *artificial*, which is to say purely mathematical, nature of geomancy that makes it an invaluable aperture onto the structure of the cosmos. That precisely the same argument was made by Francis Bacon (d. 1626) a century later, and Leibniz (d. 1716) used it as a model for modern computing, suggests Persian geomancy too as node of particular importance to the history of Western science on the one hand and the history of Pythagoreanism on the other, for all that it has been mocked or ignored in the literature to date.⁸⁰ Weirdly enough, the current AI boom has us all worshiping silicon golems Frankensteined to mathematical-linguistic life – making this a golden age of the Sand Oracle too.

The Occult Science of Empire: Imperial Grimoires for Saint-Philosopher-Kings

The Foundation of Empire is Art & Science ... Empire follows Art & Not Vice Versa as Englishmen suppose.

WILLIAM BLAKE, "On Reynolds"

Ce roi est un grand magicien.

MONTESQUIEU, *Lettres Persanes*

So much for Science; what of Empire? Here too Bacon, a father of scientific modernity and active statesman, has been held up as patron saint. He is routinely received in modern scholarship as having formulated his new-old, philhellenic and profoundly occult-scientific natural philosophy in expressly

78 Melvin-Koushki, "Occult Ecumenism."

79 Saliba, "A Sixteenth-Century Arabic Critique."

80 Melvin-Koushki, "Persianate Geomancy," pp. 187–88, 190–91; Eglash, *African Fractals*, pp. 100–1. On the garbled Latin geomantic tradition see Thorndike, *A History of Magic*, vol. 2, ch. 39. His weird mockery of the science as "a somewhat more abstruse form of superstition than the ouija board" yet mystifyingly "nearly as popular in the medieval period as the ouija board is now" (p. 110) was made over a century ago – but persists as the orthodox academic position, which refuses to recognize geomancy as integral to the history of computing due to its Afro-Islamic taint.

political terms; the natural world – and by logical extension the human – is to be *subjugated and broken*. Thus his famous declaration *nam et ipsa scientia potestas est*: ‘for knowledge itself is power.’⁸¹ And thus the prime directive, in *New Atlantis*, of his millennial Solomonic Foundation for the Enlargement of Human Empire (express model of the Royal Society): only “the Knowledge of Causes, and *Secrett Motions of Things*” can allow for “the Effecting of all Things possible.”⁸² A clearer assertion of the identity of imperialism and science, and specifically occult science, could not be desired.⁸³

This reception, however, merely perpetuates the deliberate “Enlightenment” misreading of Bacon’s project, which reduced it a prototype for colonialist triumphalism.⁸⁴ And yet it perfectly describes that of the New Brethren. Similarly cleaving to Prophet-Philosopher-King Solomon as model, the English nobleman’s Arabo-Persian peers and their sovereign Turko-Mongol patrons together strove occult-scientifically to make universal empire manifest. In a very literal sense, they therefore succeeded where Bacon could not: for the latter’s explicit aim was to rehabilitate what he imagined to be ancient *Persian* magic, as well as alchemy, astrology and various forms of divination, as the core of his new Protestantizing form of natural philosophy. He invoked magic, that is, not as superstition but as superstition’s cure.⁸⁵ That Persian cure clearly did not take in Latin Christendom, whose subsequent forms of imperialism were instead viciously antimagical. But the early modern era was nevertheless the golden age of Persian imperial magic, which greatly, scientifically aided in the enlargement of human – or rather Perso-Islamic, Turko-Mongol and Alexandrian – empire, and indeed effected all things possible to preindustrial, agrarian societies.⁸⁶

This is not to suggest that Alexander the Great’s dream of Greco-Persian world empire, abortive for over a millennium, had not been first realized by the spreading Hellenic-Abrahamic wings of the ‘Abbasid Caliphate. Yet

81 *Meditationes sacrae*, reproduced in Bacon, *The Harmony of the Essays*, p. 128.

82 *New Atlantis*, 31 (italics mine).

83 See e.g. Henry, *Knowledge Is Power*.

84 Josephson-Storm, *The Myth of Disenchantment*, pp. 44–51. Specifically, the phrase *scientia potestas est* occurs as a heresy in his *Sacred Meditations*; the full phrase was translated by Bacon or a contemporary as “for knowledge it selfe is a power whereby He knoweth.” Bacon therefore meant thereby: “*God’s power is knowledge*.” By extreme contrast, human knowledge is fundamentally weak, easily deceived and fumbling – hence the need for rigorously experimental (occult) science.

85 *Ibid.*

86 Melvin-Koushki, “Early Modern Islamicate Empire”; cf. Fleischer, “A Mediterranean Apocalypse.”

few 'Abbasid caliphs realized the specifically *occult-scientific* imperial ideal enshrined in the seminal tenth-century Pseudo-Aristotelian *Secret of Secrets* (Ar. *Sirr al-asrār*, Lat. *Secretum secretorum*), which prescribes various occult sciences – astrology, alchemy, magic and physiognomy – as primary means for achieving and maintaining Alexandrian-cum-Iranian empire.⁸⁷ But the triple apocalypse of climate change, Mongol conquest and plague between the eleventh to fourteenth centuries broke open the sociopolitical landscapes of Afro-Eurasia, making possible new globalizing forms of empire; this Great Weiriding also precipitated an occult-scientific boom of greater vigor and scope than the Western world had yet seen. The Macedonian conqueror's early modern Turko-Mongol Perso-Islamic self-styled heirs, who together achieved and maintained rule over a full third of the human race, were thus primed to pursue this peculiarly Greco-Iranian ideal more alacritously than their caliphal forebears had done, spurred on by the leading occult scientists of the post-Mongol era – to wit, Ibn Turka and his fellow New Brethren. The model of saint-philosopher-kingship the latter developed was accordingly seized upon by various Mamluk, Timurid, Ottoman, Aqquyunlu, Safavid, Uzbek, 'Adilshahi, Mughal and lesser dynasts to various degrees of success, which served in turn as a primary driver of the great cultural, intellectual and economic florescence that defined Persianate early modernity.

While this revolutionary new-old model synthesized Islamic, Chinggisid and Persian imperial inputs in classic early modern progressivist-perennialist fashion, it was above all, I argue, *Hellenizing* in tenor – whence the boom in imperial philhellenism and especially Pythagoreanism in Iran in particular during the fifteenth to seventeenth centuries,⁸⁸ and whence Ulugh Beg's status as Timurid Archytas. But I would suggest that the exemplum most relevant to and revealing of the dynamics of early modern Islamicate (occult) philosopher-kingship, though one not to my knowledge explicitly invoked, is rather that of the Roman emperor Julian the Apostate (r. 361–63): for the Hellenizing, universalist vision of this infamous would-be Alexander was openly inspired by the Syrian sage Iamblichus (d. ca. 333) and Maximus of Ephesus (d. 372), both of whom pushed a program of *Neoplatonic theurgy* to defend Hellenism against the encroachment of ascendant Christianity. Julian

87 Van Bladel, "Iranian Characteristics;" Maria Subtelny's study of its Persianate receptions is forthcoming. As that may be, the occult sciences, and especially astrology, were hardly peripheral to 'Abbasid imperial practice; see Gutas, *Greek Thought*, passim; Borrut, *Astrology and History*.

88 Again, see Terrier, "La figure de Pythagore"; Pourjavady and Schmidtke, "An Eastern Renaissance?"

failed, of course, and perished on campaign against Iran after less than three years of rule; yet he was celebrated through the Italian Renaissance and even to the present as a dangerous but entrancing figure of romance and brilliance.⁸⁹

Here Ibn Turka stands as an Iamblichus reincarnate, and his first Timurid patron, Iskandar Sulṭān (r. 1409–14), Temür's grandson and would-be successor, a Julian. The former first formulated his Pythagorean theory of lettrist sovereignty for the latter, whose short but culturally and scientifically brilliant reign in Shiraz and Isfahan is immediately reminiscent of Julian's (the latter's "pagan" revivalism aside). Like Iamblichus, biographer of Pythagoras, Ibn Turka sought to systematize Pythagorean-Platonic philosophy along strictly theurgic, Hermetic lines. (Ibn al-ʿArabī here stands in for Plotinus (d. 270) – both figures crucial to later Islamic imperial intellection.) Like Julian, Iskandar styled himself a second Alexander in occult-scientific pursuit of a Platonopolis.⁹⁰ While Ibn Turka and Iskandar largely failed in their imperial-theurgic project in their own lifetimes, also like Iamblichus and Julian, their explicitly philhellenizing and Pythagorean brand of philosopher-kingship nevertheless came to be imitated, elaborated on and sung by many of the greatest Muslim sovereigns, scholars and bards of the early modern era, and materially enshrined in mosque and mausoleum as starship.⁹¹

But much had transpired in the eleven centuries separating the two rulers – the advent of Islam most notably, that apotheosis of late antiquity.⁹² Iskandar's performance of saint-philosopher-kingship was thus predicated rather on his (Sunni-Shiʿi) identification with Imam ʿAlī and the Prophetic House, fountainhead of occult science in the Islamic dispensation, which drove his patronage of astronomer-astrologers and various other occult-scientists in turn.⁹³ Chief among them was Ibn Turka, two of whose most influential works on lettrism – *Investigations* and *On the Letters* (*Risāla-yi Ḥurūf*) – were almost certainly commissioned by Iskandar, and whose presentation of the

89 See e.g. Vidal, *Julian*; Shaw, *Hellenic Tantra*.

90 Struck, *Divination and Human Nature*, ch. 4: No mere traditionalist, Iamblichus breaks rather radically from ancient divination theory, strictly distinguishing between popular divinatory practices (oracles, hepatoscopy, oneiromancy, etc.) and true *prognosis*; while the former are merely observational natural sciences, and thus subject to error, the latter taps into nodes of divinity even in the physical realm – hence the effectiveness of theurgic ritual too. (Ironically, Iamblichus's new theory foreshadows the Christian theoretical distinction between prophecy and divination, the latter being entirely demonic.) For Ibn Turka, who makes a similarly radical intervention, these nodes are letter-number, constructs of Light.

91 Melvin-Koushi, "Mīr Dāmād's *On Doffing*."

92 Fowden, *Before and after Muḥammad*.

93 Melvin-Koushki, "Early Modern Islamic Empire," pp. 360–61.

science as quintessentially Solomonic-Pythagorean and Imamic was clearly intended to advance his patron's claim to similar status.⁹⁴ Much has been made of Iskandar's interest in astronomy-astrology in particular; he employed scientists of the caliber of Maḥmūd b. Yaḥyā Kāshī (aka 'Imād al-Munajjimīn, 'Pillar among Astrologers'), hosted debates on a variety of subjects, and tasked his book workshop (*kitābkhāna*) with the production of gorgeous, luxury compendia on the various philosophical, mathematical, literary and religious sciences.⁹⁵ It is not an accident that one of the only two Islamicate illuminated horoscopes to have reached us, and certainly the most gorgeous, is Iskandar's. Now everywhere reproduced on modern textbook and exhibit covers to represent Science in Islam (colonial consciousness is nothing if not gaslighting), it was clearly calculated to represent scientific proof of his status as philosopher-king and Lord of Conjunction (*ṣāḥib-qirān*).⁹⁶

His expansive patronage program aside, the legitimacy of Iskandar's claim to being a spiritually perfected philosopher-scientist in his own right is best seen in the preface he wrote for a comprehensive Persian manual of mathematical astronomy (*'ilm-i hay'at*), *The Sultanic Compendium* (*Jāmi'-i sulṭānī*), possibly his own work as well. In this preface the Timurid ruler provides an overview of his personal intellectual and spiritual development as proof of his status as Perfect Man (*insān-i kāmīl*) and universal ruler, emperor of realms both seen and unseen. Or as he puts it:

Inasmuch as boundless divine solicitude chose this wretch from among all His creatures and arrayed the stature of his aptitude and noble purpose with the robe of physical and spiritual caliphate, He adorned the decree of his ascendancy with the seal of *the sultan is God's vicegerent on earth* (*al-sulṭān khalīfat Allāh fī l-arḍ*) and illuminated the niche of his nature with the radiant secrets of *My saints are under My domes – none knows them but I*, making his body (*ẓāhir-ash*) the site of manifestation for the intricate virtues of kingship (*salṭanat u pādshāhī*) and his mind (*bāṭin-ash*) revelatory of intricate sciences and metaphysical knowledge.⁹⁷

94 Melvin-Koushki, "The Quest," pp. 52, 88–90, 318–20. An edition and translation of this treatise are provided in *ibid.*, pp. 463–89.

95 Binbaş, *Intellectual Networks*, pp. 89–96.

96 London, Wellcome Collection, MS Persian 474. See Caiozzo, "The Horoscope of Iskandar Sulṭān"; I riddle it in Melvin-Koushki, "Selenocentrism and Heliocentrism." On the term *ṣāḥib-qirān* see Chann, "Lord of the Auspicious Conjunction."

97 "Dibācha-yi *Jāmi'-i al-sulṭānī*," 209 (translation mine).

Most significantly, this is the first post-Mongol use of the term caliphate to refer to saint-philosopher-kingship – the ‘Abbasid model now being wholly retired, yet simultaneously restored to its original millenarian potency. The impressive Islamic universalism of Iskandar’s self-styling aside, he then claims a total mastery of “all the traditional and rational sciences in both fundamental principles and applications,”⁹⁸ including theology, philosophy and Sufism. But he finally declares astronomy-astrology (*‘ilm-i nujūm*) to be of all sciences the most useful to kings – with the exception of lettrism (*‘ilm-i ḥurūf*), the sole universal science in its status as chiefest branch of Onemaking (*‘ilm-i tawḥīd*).⁹⁹

Although he was soon foiled by his more conservative, Sunnizing uncle Shāhrukh (r. 1409–47), Iskandar, the Timurid Julian, nevertheless stands as model for the new, peculiarly early modern breed of self-theosizing and hence openly occultist Muslim rulers that prospered across the post-Mongol Persian cosmopolis as a whole. He must therefore be read as an early modern Western imperial experimentalist pursuing – quite successfully, if only after the fact – a new brand of millennial, monist-universalist sovereignty predicated on an occult-scientific understanding, a *deep reading*, of history and the cosmos.

But this deep reading, this imperial ontogrammatology, first requires *reading*. Breaking with long scholarly precedent, whereby occult-scientific knowledge must be rendered illegible to noninitiates, Ibn Turka accordingly inaugurated a new *imperial grimoire* culture by penning a series of attractive and accessible Persian treatises on lettrism for his Timurid patrons, beginning with Iskandar – a model that would be eagerly embraced by a host of persophone would-be imperial ideologues over the next several centuries. In effect, this move represents the epochal and definitive *de-esotericization*, and indeed *textualization*, *popularization* and *democratization*, of occultism – all developments quintessentially early modern in tenor. Lettrist treatises circulated during the twelfth and thirteenth centuries in “esotericist reading communities” fiercely protective of their secrets;¹⁰⁰ but from the late fourteenth onward, due to the efforts of Ibn Turka and Bistāmī in particular, and exploiting the sociopolitical vacuum left by the ravages of that century, they increasingly circulated among and were targeted at court elites, while being open to anyone conversant in Persian or Arabic.¹⁰¹ *History of science*, in short, requires *history of empire*; and history of Persianate occult-scientific empire is *history of the book*.

98 Ibid.

99 Ibid., p. 210.

100 Gardiner, “Esotericism in a Manuscript Culture.”

101 Melvin-Koushki, “*How to Rule the World*”; idem, “The Occult Court.”

Space does not permit a survey of this Persian imperial grimoire culture here, almost wholly ignored by early modern intellectual, social and political historians to date. But three of Ibn Turka's works are here representative of the flavor of that burgeoning culture. Mention has already been made of *On the Letters*, written for Iskandar in 1414, and referenced by the latter in his preface to *The Sultanic Compendium*. (The *Sharḥ al-basmala*, dedicated to Ulugh Beg, as noted, is neither in Persian nor a grimoire, for all that it was meant to perform a similar imperial function.) With Iskandar's defeat and execution at the hands of Shāhrukh, ultimate victor in the succession struggle that followed Temür's death, Ibn Turka sought to cultivate his patronage in turn – by means of, naturally, an explicitly imperial manual of lettrism, *Query of Kings* (*Su'l al-mulūk*). As the title suggests, this highly accessible introductory treatise, written for Shāhrukh in early 1426 at the instance of the latter's son Bāysunghur (d. 1434), a major patron of the book arts and accomplished calligrapher in his own right, represents a catalogue of various forms of political divination and political magic, and treats of both theoretical and applied (i.e., alchemical and talismanic) lettrism, developing a compelling theory of the mind-body relationship to this end. Its central theme: the sovereign as Jesus-like *physician of empire*.¹⁰² Most notably, the *Query of Kings* ends with a lettrist prognostication as to the duration of the Timurid – that is to say, Shahrukhid – state, whereby Ibn Turka acclaims Iskandar's uncle the fated ruler and political model of the ninth Islamic century as a whole.¹⁰³

Ironically, our Isfahan lettrist's prognostication was to prove all too accurate: Shāhrukh's more classic Sunni model won out over Iskandar's radical, Pythagoreanizing program, at least for the nonce, and Ibn Turka died a frustrated exile. Yet the latter's countermodel was to prove far more attractive during the following century, as ambitious rulers and Sufi messiahs competed fiercely for sacral power (*walāya*) in the runup to the Islamic millennium (1592 CE).

Here Ibn Turka's last work written for Shāhrukh, similarly at the instance of his prince-calligrapher patron Bāysunghur, is especially salient: therein he formulates his lettrist theory of Islamic universalist imperialism most explicitly and succinctly. Completed in 1426, the *Debate of Feast and Fight* (*Munāzara-yi bazm u razm*) is an ornate Persian work that expressly imperializes the

102 On Jesus as model for Mughal and Ottoman imperialism alike see Melvin-Koushki, "Early Modern Islamic Empire"; idem, "Selenocentrism and Heliocentrism."

103 This he does through a straightforward analysis of Shāhrukh's name, which converts to *shāh-i rukh*, i.e., *shāh-i RKh*, or 'shah of 900.' See Melvin-Koushki, "The Quest," pp. 311–14; an edition and translation of this treatise is provided in idem, *Persian Pythagoreanism*.

venerable feast vs. fight (i.e., court vs. military, or pen vs. sword) trope within a lettrist-literary framework. For the first time in the centuries-old Arabo-Persian *munāẓara* tradition, which resisted any debate's resolution, Ibn Turka *marries the opposites* in a Pythagorean mode clearly meant to be instructive to his Timurid royal patron: he is to perform the role of Emperor Love (*sultān 'ishq*), transcendent of all political-legal dualities, and harmonizing the same, thereby bringing the body politic back into the balance that is perfect health. In high Pythagorean fashion, in short, he argues for the Timurid Empire as ineluctable historical manifestation of the One. This lettrist mirror for princes is thus not simply unprecedented in Persian literature, a typical expression of the ornate literary panache of these sciencers of letters, but also serves as key to Timurid and post-Timurid universalist imperial ideology itself in its formative, experimentalist phase.¹⁰⁴

Ibn Turka's vision for his new brand of occult philosopher-kingship may be seen, in sum, as a composite of his three Timurid patrons, Iskandar, Ulugh Beg and Bāysunghur: an Alexandrian, world-conquering patron of the arts and sciences – especially the occult sciences – and a scientist and an artist in his own right, willing to take lettrism as master guiding principle in the theoretical and physical construction of henological empire. In this project, moreover, he was rather more successful than Plato himself, who failed to turn the Sicilian tyrant Dionysius II (r. 367–56, 347–44 BCE) into a philosopher-king; but Doctor Littleturk induced these Turkish warlords to make of Iran a true Platonopolis, or rather *Pythagoropolis* – by means of Persian magic.

Riddling and Poeticizing, Painting and Building, the Superreal World

a wise and letters play of all you can ceive

JAMES JOYCE, *Finnegans Wake*

The book of nature, like the book of scripture, demands interpretation: the world must be mathematically-linguistically *riddled*. For the New Brethren, this was no empty metaphor, but one pregnant with both occult-scientific and aesthetic possibility.

It is thus no accident that Yazdī, extending Ibn Turka's project, was the first in the Arabo-Persian literary tradition – and indeed the Western tradition as

104 For an analysis and partial translation see Melvin-Koushki, "Imperial Talismanic Love"; an edition and full translation of this work is likewise provided in my *Persian Pythagoreanism*.

a whole – to explicitly theorize the humble logograph or riddle (*mu'ammā*) as a crucial skill in the lettrist's technical repertoire: an immediate, poetic-mathematical means of analyzing a person's name in order to discern their character, perhaps even their fate.¹⁰⁵ (Similarly, chronograms, properly constructed, offer insight into the texture of history; and in Jābirian and Jildakian Arabic alchemy, to analyze the name of a substance is to know its elemental essence.¹⁰⁶) Logographs came to be most commonly deployed as social calling cards, to be sure; but their extreme popularity in Persianate culture from the fifteenth century onward testifies to a broader social consciousness, informed by our influential Timurid lettrists, that the world is *semantic*, and hence deconstructable – and reconstructable – at a formal level. The same observation may be extended to contemporary Mamluk Arabic literary culture, wherein a preoccupation with the formal also prevailed as expression of a general Mamluk “linguistic consciousness” that achieved the “poetization of everyday life.”¹⁰⁷ Yazdī's argument for the appearance of the *mu'ammā* genre itself as harbinger of a new age would thus seem to have been justified by subsequent developments.¹⁰⁸

Our lettrist historian's move, it must be stressed, is both unprecedented in the Western literary tradition and a natural apotheosis of the same. For isopsephy or *gematria* (Hebrew calque on Greek *geometria*) – the technique that was to emerge, a millennium later, as a mainstay of lettrist-kabbalist practice, and thence the basis of the *mu'ammā* – was first developed for and employed in late antique Greek riddle culture; here Leonides (fl. 1st c. CE) was the pioneer.¹⁰⁹ Yet the Alexandrian riddler did not explicitly couch his isopsephic art in a Pythagorean cosmological framework. This is in contrast to Valentinian number theology of the next century, which was taken up by various Gnostic movements in turn, whence it entered (“extremist” Shi'i) Islam in its formative phase; Greek arithmosophy thus became a primary basis for the subsequent

105 Melvin-Koushki, “The Quest,” pp. 379–89; Binbaş, *Intellectual Networks*, pp. 48, 81–89. This he does in his seminal treatise on the subject, *Embroidered Robes* (*Hulal-i muṭarraz*).

106 Melvin-Koushki, “The Quest,” pp. 180–82; Harris, “Better Religion through Chemistry.”

107 Bauer, “Mamluk Literature,” pp. 109, 130. Cf. the simultaneous rise of the ‘artificial ode’ (*qaṣīda-yi maṣnū'*) genre, long cited, with the *mu'ammā*, as proof of Timurid-Turkmen cultural decadence; but Paul Losensky in particular has shown them to rather epitomize the period's structuralist-textualist turn, bent on the codification and amplification of the whole of the Persian poetic tradition (*Welcoming Fighānī*, pp. 154–64).

108 Melvin-Koushki, “The Quest,” pp. 327–28, 385.

109 Page, *Further Greek Epigrams*, pp. 503–41. The Number of the Beast in the Book of Revelation – 666, or 616, perhaps encoding Nero's (r. 54–68) name – is another early instance.

development of Hebrew kabbalah and Arabic lettrism alike.¹¹⁰ But the fusion of lettrism and literature – the scientific reading and writing of *form as content* – was a project peculiar to the New Brethren, who helped make the riddle sociopolitically *de rigueur*.

As for the linguistic consciousness of these scientists of letters and literateurs more generally: Yazdī was responsible for one of the most widely imitated ornate Persian chronicles ever written (on which more below), and an accomplished poet; Bisṭāmī penned an important manual of Arabic epistolary style;¹¹¹ and Ibn Turka himself was hailed by the literary historian and Pahlavi poet-laureate Muḥammad-Taqī Bahār (d. 1951) as the greatest stylist of ornate Persian prose of the fifteenth century, indeed the first writer to marry literary (*adabī*) style to scientific (*ilmī*) content.¹¹²

Aesthetics was thus of paramount concern to the New Brethren, who argued, in high Pythagorean fashion, for the beautification of the world scientifically, literarily and hence imperially, with no little success. That is to say: If the world is an imperial text, then that text must needs be *calligraphic*, and the culture that writes and reads it must needs be *lettrary*.

It is therefore no coincidence that, in tandem with the starry arts, patronage of the arts of the book boomed under the Timurids – just as Ibn Turka was formulating the first systematic Arabo-Persian metaphysics of writing. Responding to this cultural transformation, by the end of the Timurid period historians began to pay far more attention to calligraphers and painters than had ever before been merited; and in the sixteenth century, under the successor Safavids, an entirely new art-historical genre was born: the album preface.¹¹³ This genre is naturally of primary importance for understanding Timurid-Safavid writerly-artistic culture, and has been celebrated by Islamic art historians as such. But they have thus far failed to take seriously the often expressly *psychedelic* tenor of that genre.¹¹⁴

Here too Ibn Turka's ontogrammatology provides a key. As I argue, his unprecedented declaration of the epistemological-ontological superiority of sight to hearing, on strictly lettrist grounds, provided a theoretical basis for the burgeoning of Persianate book culture and visual culture more broadly. That Bāysunghur, his second Timurid patron, and commissioner of two of

110 I here extend the arguments of Acevedo, Kalvesmaki and Wasserstrom to the Arabic context.

111 Scil., *Methods of Solicitation: On the Pleasures of Letterwriting* (*Kitāb Manāhij al-tawassul fī mabāhij al-tarassul*).

112 *Sabk-shināsī*, vol. 3, pp. 223–34, 352–53; Melvin-Koushki, “The Quest,” pp. 379–407.

113 On which see Roxburgh, *Prefacing the Image*, pp. 125, et passim.

114 Melvin-Koushki, “Of Islamic Grammatology.”

his most important imperial lettrist treatises, achieved renown precisely as a calligrapher has not a little to do with Ibn Turka's valorization of the quranic category *men of main and vision* (*ūlū l-aydī wa-l-abṣār*, Q 38:45) over *those of verbal remembrance* (*ahl al-dhikr*, Q 16:43, 21:7) – just as his valorization of *men possessed of minds* (*ūlū l-albāb*, Q 2:179, etc.) explicitly celebrates the mathematical labors of Ulugh Beg. As for painting, consistently theorized in Persian as an extension of calligraphy: our Isfahani occult philosopher's *tashkīk al-ḥarf* schema explicitly structures the most famous Persianate theorization and defense thereof, that of Abū l-Faḥr 'Allāmī (d. 1602), chief architect of the Mughal emperor Akbar's (r. 1556–1605) Indian empire, who accordingly asserted the ontological and epistemological superiority of Sight to Speech in terms of a lettrist-emanationist metaphysics-physics of Light.¹¹⁵

As art historians have long pointed out, this new early modern Persianate painting tradition is immediately comparable in its sheer ebullience to contemporary Italian Renaissance developments. Yet, sadly, it never managed to achieve the *perspectivism* or *naturalism* so definitive of the latter, but remained doggedly medieval – or so the old Orientalist chestnut goes. As Gülru Necipoğlu has shown, however, Persianate painting embraced in its stead *superrealism*, that is, a privileging of cosmic realism over optic realism, of imaginal insight over physical sight.¹¹⁶ (Naturalistic and perspectivist elements were hardly absent, but merely subordinated.) Latinate and Persianate painting are thus simply two paths that diverged in an early modern Western wood, two equally robust answers to the same theocentric cosmological question.

But which Western approach, the perspectivist or superrealist, was the more properly *Pythagorean*? The Italian Renaissance, of course, was precisely a renaissance of Pythagoreanism; thus numerous studies have shown perspectivism to have been often expressly theorized as an application of the same. In particular, the *linee occulte* used to construct the illusion of three dimensions on a two-dimensional surface were conceived by some to be actually *occult*, in the scientific sense of that term.¹¹⁷ But my sources make clear that superrealism too was consciously pursued as a Pythagorean discipline to at least the

115 Ibid., pp. 91–93. The shy phenomenon of sonoluminescence – Tricky and psychedelic – would seem to corroborate Ibn Turka's vision-forward philosophy, whereby even sound can be experienced as but wavelengths of Light; see Domnitch and Gelfand, "Hyperspectral Percipience."

116 On the Neoplatonic, Aristotelian and Sufi discourses increasingly used to celebrate and promote this visual culture see Necipoğlu, "The Scrutinizing Gaze."

117 Hersey, *Pythagorean Palaces*, pp. 84–87, esp. 194: "Thus a perspective construction is a way of visualizing a metempsychosis. In this and other ways *linee occulte* are truly occult: they manifest or even create hidden links. Among these links are astrological influences and

same extent. We must therefore henceforth treat Latinate perspectivism and Persianate superrealism as two sides of the same Pythagorean coin, whereby *vision-as-mathematical-construct* finally theoretically and practically trumped oral culture for the first time in the Western tradition – whence the burgeoning book and visual cultures of Greater Western early modernity.

(Parenthetically, this is not to suggest that the valorization rather of hearing over sight in the post-Mongol Persianate context could not be motivated by the same Pythagorean-occultist commitment; the Old Brethren, after all, did exactly this. Thus *music* – that central Pythagorean science, whence its place in the quadrivium – came to be theorized and practiced in early modern Persianate India as an important astrologico-magico-medical means of *healing and transforming society*.¹¹⁸)

What of architecture, the chiefest means of materially calligraphing empire? Due to an often essentializing and ahistorical identification of Pythagoreanism with “sacred geometry” in the literature, many Islamic architectural historians have understandably shied away from emphasizing this distinct philosophical current as a likely theoretical and especially sociopolitical framework for the construction of various celebrated Timurid, Safavid, Mughal, Uzbek or Ottoman imperial monuments.¹¹⁹ But this reticence is unfortunate, as it serves but to further Un-West Islamdom; it is both artifact and index of the dire state of early modern Islamic intellectual and imperial history, which remains in its infancy, and hamstrung by persistent occultophobia. Again, the present study is the first in the field to treat of this, the golden age of Islamic Pythagoreanism. Given how robustly parallel Persianate and Latinate developments ran (culturally if not imperially) during the Renaissances of Western early modernity,

gravity, which is ... the architectural form of love.” On the latter Pythagorean theme see Melvin-Koushki, “Imperial Talismanic Love.”

118 Schofield, “Music, Art and Power.”

119 My thanks to Professor Necipoğlu for this observation. Exemplifying such ahistoricism is Seyyed Hossein Nasr’s formulation as follows, which nevertheless links, correctly, Pythagoreanism with lettrism (*Islamic Art and Spirituality*, p. 47): “There is within Islamic spirituality a special link with qualitative mathematics in the Pythagorean sense ... [But] the mathematical nature of Islamic art and architecture does not derive from external historical influences, Greek or otherwise. It derives from the Quran whose own mathematical structure is bewildering and reveals an amazing rapport between Islamic intellectual and spiritual concerns and mathematics. The mathematical nature of Islamic art is in a sense the externalization of the mathematics hidden in the very structure of the Quran and the numerical symbolism of its letters and words. The Pythagorean philosophy of mathematics provided the language and presented an already elaborated science, itself of an esoteric nature and going back to Egypt and Babylon, for the ‘spiritual mathematics’ which is so central to Islamic architecture and even the so-called decorative arts.”

however, and given the increasing (though uneven and partial) entrenchment of Pythagorean-lettrist sensibilities in philosophical, scientific, literary and painterly circles of the era, I propose monumental architecture too as a crucial node of comparative historical study going forward.

For many Latinate thinkers and doers, to architect is to be a second Pythagoras. A few went farther to declare the discipline an occult science. Marsilio Ficino (d. 1499) himself, that great Italian reviver of Neoplatonism, “held that buildings possessed magical powers and that their architects were demiurges.”¹²⁰ The fact that architecture was first textualized and mathematized in Timurid Iran, several of whose monuments do not merely express imperial universalism in a “baroque” mode but strain toward infinity through Penrose tiling,¹²¹ is therefore most telling in this lettrary context. Most notably, the emphasis in Timurid architecture on geometrical order, hierarchy, proportion, axiality, symmetry, harmony and infinity – all classic Pythagorean ideals – has invited comparison to a similar emphasis, for example, in the great Renaissance architects Filippo Brunelleschi (d. 1446) and Leon Battista Alberti (d. 1472).¹²² But the latter were passionate, card-carrying Pythagoreans, as a number of studies have shown;¹²³ why should we not expect a similar motivation of their Timurid peers?

And as with Ibn Turka’s legacy itself, the Timurid quest for architectural harmony in the fifteenth century only intensified in the sixteenth and seventeenth, as their Safavid, Uzbek and Ottoman rivals sought to outdo them and one another in the construction of imperial space.¹²⁴ The interior of the Shaykh Luṭf Allāh mosque of the new Safavid capital of Isfahan, for example, just off its central Map of the World (*naqsh-i jahān*) maydan, visually and structurally encodes the Ibn Turkian *tashkīk al-ḥarf* schema to an unprecedented and perhaps unparalleled extent: number-letters of light radiating from the One of the central dome to mathematically and then linguistically constellate the Many (see Plate 1). That a number of the accompanying poetic inscriptions were composed by Shaykh Bahā’ī (d. 1621) – the Safavid Chief Jurisconsult (*shaykh al-islām*), philosopher, mathematician, astronomer and architect who is famed to this day as the most powerful practicing mage of his generation, and student in lettrism of the New Brother Maḥmūd Dihdār – is

120 Hersey, *Pythagorean Palaces*, p. 192.

121 Bernard O’Kane, *Timurid Architecture in Khurasan*, p. 110. See also Lu and Steinhart, “Decagonal and Quasi-Crystalline Tilings”; my thanks to Michael Noble for this reference.

122 Bloom, review of Golombek and Wilber, *Timurid Architecture*.

123 E.g., Joost-Gaugier, *Pythagoras*; Hersey, *Pythagorean Palaces*.

124 On this intra-Persianate imperial-architectural contest see Necipoğlu, “Transregional Connections.”



PLATE 1 Interior, Shaykh Lutf Allāh Mosque, Isfahan (Kazuyoshi Nomachi)

likewise suggestive, as is the fact that the exterior dome is psychotropically adorned with the divine Names, centerpiece of lettrist theory and practice. Indeed, the same sage-mage was responsible for the revolutionary redesign of Isfahan as a whole, which rendered the city both a model of paradise and a transhumanist machine by perspectivist means: urban planning as a form of geometrical poetry-as-sorcery.¹²⁵ A fitting imperial fate for our occult philosopher's hometown!

But it was Temür's successors in India who took the palm in this imperial Pythagorean contest. Given the rudimentary state of the field, however, two instances must here suffice. The second Mughal emperor Humāyūn (r. 1530–40, 1555–56), like his Safavid rival Ṭahmāsb (in)famous for his dedication to the occult sciences, erected in his Agra palace complex a Talisman House (*khāna-yi ṭilism*) wherein he conducted court ritual and even entertainment in emphatically astral-magical terms – a practice his more glorious son Akbar would continue with greater pomp.¹²⁶ (Ficino would have been delighted.)

125 Melvin-Koushki, "Mīr Dāmād's *On Doffing*"; Gharipour, "Gardens of Safavid Iran"; Emami, *Isfahan*; idem, "Discursive Images"; idem, "Religious Architecture."

126 Koch, *The Planetary King*. On Mughal astral-magical court ritual generally see Orthmann, "Court Culture"; Moin, *The Millennial Sovereign*; Truschke, "Translating the Solar Cosmology."

And talismans – and talismanic cities and their talismanic sovereigns – aside, the specifically Timurid imperial lettrary culture constructed by Ibn Turka and Yazdī was finally fully bodied forth in three dimensions by Emperor Shāhjahān's (r. 1628–57) Tāj Maḥall, the greatest achievement of Indo-Timurid architecture: for this mathematical model of paradise and shrine to all-creative Light was precisely designed to be both cosmologically riddled and ridden as a starship. The still popular story of Shāhjahān's unrealized plan to build a Black Tāj for himself on the site of Moonlight Garden across the Yamuna River to mirror the White for Mumtāz as testament to their terrestrial-celestial Love is almost certainly a European and especially British Romantic invention; yet it too encodes a curiously similar talismanic testament to the *coincidentia oppositorum* as Turko-Mongol Perso-Islamic imperial ethos.¹²⁷

History Becomes a Pythagorean Science

When one reads these strange pages of one long gone one feels that one is at one with one who once....

JAMES JOYCE, *Ulysses*

The post-Mongol Persianate world, then, was often defined by Islam as Magic, Magic as Science, Science as Empire. Imperial grimoires accordingly proliferated, and some imperial spaces were explicitly constructed as talismans, lettrist riddles and/or theurgic starships. But how to write the history of this Islamic occult-scientific empire?

By Pythagoreanizing historiography too: for history is a story is a spell – and so easily formally transformed into a technology of time.

127 Parodi, “The Distilled Essence”; Koch, *The Complete Taj Mahal*. Likewise suggestive is the fact that the two sides of Mumtāz Maḥall's (d. 1631) lower cenotaph feature the 99 Names of God in their entirety (Begley and Desai, *Taj Mahal*, pp. 240–41) – the primary focus, of course, of lettrist magic by the early modern period. More broadly, as Ebba Koch notes, the Tāj Maḥall “is ‘built architectural theory,’ which can be read almost like a literary text once we have mastered the grammar and vocabulary of the architectural language ... We note here the purest expression of a consistent formal systematization characteristic of the entire art of Shah Jahan; it represents a distinctive and outstanding contribution specific to this period” (“The Taj Mahal,” pp. 137–38). For its cosmotechnic context see Melvin-Koushki, “Selenocentrism and Heliocentrism.” On the similarly revolutionary nature of Mughal garden culture, with comparisons to Renaissance Europe, see Koch, “Carved Pools”; cf. Subtelny, *Le monde est un jardin*.

Such was the achievement, not surprisingly, of Yazdī, who won his greatest fame as dynastic historian to the Timurids; his *Ẓafarnāma*, an ornate Persian history of Temūr completed in 1436, was much imitated for centuries thereafter throughout the Persianate world.¹²⁸ What has not yet been noted by specialists, however, is the fact that Yazdī's *Book of Conquest* is predicated on a new, expressly occult-philosophical theory of history, wherein lettrism and conjunction astrology serve not as sources for mere rhetorical embellishment but rather as macrohistorical analytical framework, indispensable tools for penetrating past the phenomenal surface of historical events to get at their cosmic meaning, for cracking the code of cyclical time.¹²⁹ In other words: Yazdī here imports and extends Ibn Turka's signature dual lettrist-astrological platform for Timurid universalist imperialism to make the discipline of history an occult science.

Without Islamic and indeed Western precedent, such a daring move should be seen, I suggest, as part and parcel of a larger Arabo-Persian scholarly trend during the fifteenth and sixteenth centuries toward reformulating history as a proper and rigorous science, even as supreme among the literary sciences.¹³⁰ It is hardly an accident that this new impetus to theorize imperial historiography coincided with the great burgeoning of Pythagoreanism at the hands of the New Brethren, who stand as some of the most influential imperial ideologues of the fifteenth century; and Yazdī was the first to apply the Brethren's doctrine to the science of history.¹³¹

This he did, I have elsewhere argued, in direct rebuttal of Ibn Khaldūn's 'science of civilization' (*ʿilm al-ʿumrān*), predicated on a simplistic theory of dynastic cycling, which superficially resembles Yazdī's own cyclical theory of history. (It must again be emphasized that the two historians were both members of Barqūq's court in Cairo.) But where the latter deploys lettrism and astrology within a dual-aspect monist framework to trace the deeper patterns

128 Yazdī also began four other chronicles, including a *Muqaddima*, *Fathnāma-yi Šāhib-qirānī*, *Fathnāma-yi Humāyūn* and the *Second Maqāla*, none of which he finished and whose shifting ideological strategies reflect the displacement of Temūr's dispensation with that of his son Shāhrukh; see Binbaş, *Intellectual Networks*, pp. 199–250.

129 Melvin-Koushki, "In Defense of Geomancy"; idem, "Toward a Neopythagorean Historiography."

130 Markiewicz, "History as Science." Cf. Giambattista Vico's (d. 1744) historiographical marriage of philology and philosophy in his similarly cyclicist *New Science*.

131 The Old Brethren of Purity, to be sure, similarly formulated a Neopythagorean, cyclical theory of history thoroughly astrological in tenor; but unlike their fifteenth-century heirs, they left lettrism to the Sufis, and operated in strict anonymity – that is to say, they certainly did not serve as *imperial lettrist historians* in the manner of Yazdī.

of cosmic time, robustly theorizing *imperial transcendence* as a necessary historical expression of the One, the strictly dualist leveling theory proposed by his Tunisian colleague and rival, a rabid anti-occultist, rather condemns dynasts to be forever ground under by the cycling of history between two sociological poles. Ibn Khaldūn simply moralizes, imprisoning political actors in the binaries of time and the strictures of Shari'a; he offers in return only the humble rewards that accrue through the strict enforcement of piety on an always erring society. But Yazdī promises to historically empower and spiritually perfect – to *divinize* – its royal patrons by means of Akbarian occult science. Not only can the future be read by the aspiring (occult) philosopher-king; it can also be *rewritten*.¹³²

The latter prospect could not but be far more attractive to the ambitious millennial sovereigns of the post-Mongol era.¹³³ Ibn Khaldūn's depressing, stridently anti-millenarian, anti-monist and anti-occultist 'science of civilization' was thus roundly ignored by scholars until the middle of the more sedate seventeenth century, when Ottoman historians first began to cite it in support of their own theories of dynastic cycling.¹³⁴ Even the Mamluk historian Taqī l-Dīn al-Maqrīzī (d. 1442), Ibn Khaldūn's student and sole defender, and an enthusiastic occultist nevertheless, sought to combine his mentor's model with the Yazdian.¹³⁵

But the precedent Yazdī set was irreducibly *Timurid*. Most notably, he is responsible for the definitive Timuridization of the astrological title Lord of Conjunction (*ṣāhib-qirān*) – this by means of *lettrist* argument.¹³⁶ That is to say, he is the first historian in the Arabo-Persian tradition to construct a dual astrological-lettrist platform in support of his patrons' imperial legitimacy, whereby the analysis of a dynast's horoscope must be paired with the lettrist analysis of names and events, especially as they relate to quranic passages, and particularly the cryptic disconnected letters opening certain suras (*muqatta'āt*). (Ibn Turka's *Query of Kings*, it will be recalled, demonstrated the imperial utility of this science for the benefit of Shāhrukh.) Thus Temür, in Yazdī's *deep reading* of history, is not only supreme Muslim Lord of Conjunction, preordained world conqueror and second Alexander; he is also an epoch-making historical

132 Melvin-Koushki, "In Defense of Geomancy."

133 On the category of millennial sovereignty in the Timurid-Mughal context see Moin, *The Millennial Sovereign*.

134 Fleischer, "Royal Authority," p. 199.

135 As Robert Irwin summarizes al-Maqrīzī's historiographical project ("Al-Maqrīzī and Ibn Khaldūn," 230): "History was at the core of his oeuvre, but occult and eschatological concerns were at the core of his history."

136 Binbaş, *Intellectual Networks*, 255; Melvin-Koushki, "In Defense of Geomancy."

manifestation of both the *coincidentia oppositorum* – primary principle of Ibn Turka's lettrist metaphysics – and the quranic *ALM*.¹³⁷

Any direct historiographical challenge to this new, scientific Timurid vocabulary of sovereignty would henceforth require the counterdeployment of astrological and lettrist arguments in tandem. Such a challenge was accordingly mounted by some of the most prominent scholar-ideologues and Persian historians through at least the seventeenth century, as the great Turko-Mongol Perso-Islamic empires jostled for supremacy within Islamdom and the Afro-Eurasian ecumene as a whole. These include in the first place Jalāl al-Dīn Davānī (d. 1502), famed Shirazi philosopher and Ibn Turkian lettrist,¹³⁸ and two of his students, court secretary-historians in the Aqqyunlu chancery: Fazl Allāh Khunji Iṣfahānī (d. 1521) and Idrīs Bidlīsī (d. 1520). (Both men, significantly, had themselves sojourned in Cairo.) As an opening salvo, Davānī sought to sidestep and undercut Yazdī's lettrist arguments in the *Book of Conquest* by seizing upon the opening verses of Sūrat al-Rūm ('The Byzantines,' Q 30:1–5),¹³⁹ the only explicit political prediction in the Quran, to prove the cosmic superiority the Aqqyunlu sultan Uzun Ḥasan (r. 1457–78) to Temür, and hence the historical inevitability of Aqqyunlu imperial supremacy within Islamdom; this became the linchpin of Aqqyunlu ideology as expressed in diplomatic correspondence and historiography thenceforward.

Thus Khunji's *Aminian World-adorning History* (*Tārīkh-i Ālam-ārā-yi Amīnī*), an ornate Persian history of the dynasty expressly on the model of Yazdī's, adds to Davānī's lettrist counterargument others of a similarly occult-scientific tenor, as well as innovative legal arguments in support of the Aqqyunlu dynasts' status as messianic centenary 'renewers' (sg. *mujaddid*) – a venerable title Yazdī had likewise helped politicize and Timuridize.¹⁴⁰ And thus the *Eight Paradises* (*Hasht bihisht*) of Bidlīsī, Aqqyunlu turncoat, was the first comprehensive Persian history of the Ottoman dynasty, and commissioned by Sultan Bāyezīd II (r. 1481–1512) himself, that great institutionalizer of

137 Yazdī, *Ẓafarnāma*, vp. 1, pp. 236–38, 404–5, vol. 2, p. 1295; these passages are translated in Melvin-Koushki, "In Defense of Geomancy," pp. 388–91. The letter sequence *ALM* opens suras 2, 3, 29, 30, 31 and 32.

138 On Davānī as lettrist see Melvin-Koushki, "The Quest," pp. 247–61; idem, "World as (Arabic) Text."

139 *ALM*. *The Byzantines* (al-Rūm) *have been vanquished in the nearer part of the land; and after their vanquishing they shall vanquish in a few years. God's is the Command before and after, and on that day shall the believers rejoice in God's help. God helps whomsoever He will; He is the Almighty, the All-compassionate.*

140 Woods, *The Aqqyunlu*, pp. 102–6, 145.

astrology.¹⁴¹ Responding to Yazdī's and Khunjī's histories in particular, Bidlīsī's is a gorgeous and highly complex work of historiography that focuses on the "cosmically ordained and universally applicable signs of Ottoman superiority," drawing upon "a wide array of astrological, mystical, philosophical, and occult doctrines and [freely mixing] ideas from these varied traditions to create a compelling and universalizing conception of kingship."¹⁴² To this end, for example, it ontologically equates – in unmistakably Pythagorean fashion – Bāyezīd with the eighth and highest celestial sphere, where the totality of existents first take shape. And that Ottoman sovereign, not Temür, is the greatest Lord of Conjunction of the last age – a point easily proved by an analysis of Bāyezīd's horoscope.¹⁴³ Bidlīsī also pointedly retools Davānī's and Khunjī's lettrist argument for Ottoman purposes: the Sūrat al-Rūm prophecy, he shows at length, cannot refer to Uzun Ḥasan's defeat of the Timurid Abū Sa'īd (r. 1451–69) at the Battle of Muş in 1467, but rather to Fatih Meḥmed's (r. 1444–46, 1451–81) victory over Uzun Ḥasan at Otlukbeli in 1473.¹⁴⁴

The *Eight Paradises*, in short, represents a masterful and polemical Ottomanization of the Timurid-Aqqayunlu tradition of Pythagorean-occultist historiography for the benefit of his rival patron. This bold Persian masterwork became in turn the model for the new, far more ornate mode of Ottoman historiography in Turkish that became standard thereafter, including most notably the *Histories of the Ottoman House* (*Tevārīḥ-i āl-i 'Osmān*) of Kemālpaşazāde (d. 1534), Ottoman Chief Jurisconsult, which inherited Yazdī's cosmic and occult-scientific orientation, if only in part.¹⁴⁵

The foremost Ottoman member of the New Brethren provided here the imprimatur. In his autobiographical *Pearl in the Diadem of Tractates* (*Durrat tāj al-rasā'il*), Ibn Turka's fellow Cairene sojourner Bisṭāmī, preeminent source of Ottoman imperial lettrism, proposes a startlingly new classification for the discipline of history (*'ilm al-akhbār wa-l-siyar*): he moves it from the traditional (*naqlī*) sciences – where every other Arabic and Persian encyclopedist had classed it to date – to the rational (*'aqlī*) sciences, and specifically the

141 Şen, "Astrology in the Service of the Empire"; and see now idem, *Forgotten Experts*.

142 Markiewicz, "The Crisis," pp. 147, 342. The latter quote refers in Markiewicz's study rather to the unprecedented eclecticism of the Timurid model, here exemplified by Yazdī and emulated by Khunjī and Bidlīsī. And see now Markiewicz, *The Crisis*.

143 Bidlīsī, *Hasht Biḥisht*, Süleymaniye MS Nuruosmaniye 3209, ff. 497b, 498b; Markiewicz, "The Crisis," pp. 378–79; Şen, "Reading the Stars."

144 Bidlīsī, *Hasht Biḥisht*, Süleymaniye MS Nuruosmaniye 3209, f. 470b; Markiewicz, "The Crisis," p. 337.

145 Markiewicz, "The Crisis," pp. 138–40; Şen, "Practicing Astral Magic," pp. 85–86; idem, "Reading the Stars;" Melvin-Koushki, "Toward a Neopythagorean Historiography."

applied mathematical sciences. Equally tellingly, he there allies history with the occult sciences of bibliomancy (*‘ilm al-fā’l*), magic and amuletry (*‘ilm al-siḥr wa-l-‘azā’im*) and alchemy and engineering (*‘ilm al-kīmīyā’ wa-l-ḥiyāl*), on the one hand, and the linguistic sciences of prosody (*‘ilm al-‘arūd*), lexicography (*‘ilm al-lughā*), grammar (*‘ilm al-naḥw*) and writing (*‘ilm al-kitāba*), on the other, as well as economics (*‘ilm al-bay’ wa-l-shirā*).¹⁴⁶ Nor was this a mere theoretical exercise: this Ottoman New Brother also wrote a universal chronicle, *Regulation of Conduct: On the Edification of Kings* (*Naẓm al-sulūk fī musāmarat al-mulūk*), to demonstrate how his occult-scientific, Sufi and millenarian project should be put into imperial historiographical practice.¹⁴⁷ To thus “mathematize” history, to posit its object as linguistic construct in high lettrist fashion, is to argue for the discipline’s status as applied Pythagorean science.¹⁴⁸

Safavid historians, needless to say, did not let this scientific-historiographic challenge go unanswered; lettrist and astrological analyses would seem to suffuse Safavid imperial historiography to a similar extent. Siyāqī Nizāmī’s (d. 1603) *Imperial Conquests* (*Futūḥāt-i humāyūn*), for example, attempts to prove Shah ‘Abbās’s status as a Lord of Conjunction ontologically equivalent to Temür, Chinggis Khan and Alexander – purely on the basis of lettrist analysis.¹⁴⁹ In his *History of ‘Abbās* (*Tārīkh-i ‘Abbāsī*), the same shah’s chief court astrologer, Jalāl al-Dīn Munajjim Yazdī (d. 1619) – author of an astrological and a geomantic manual both –, unsurprisingly prefers astrological explanations over lettrist.¹⁵⁰ Yet this historian-occultist was also responsible for the seminal Safavid expansion of the Timurid polymath Ḥusayn Vā‘iz Kāshifī’s (d. 1505) grimoire *Qasimian Secrets* (*Asrār-i Qāsimī*), constellating a who’s who

146 *Durrat tāj al-rasā’il*, Istanbul, MS Nuruosmaniye 4905, f. 20b; my thanks to Cornell Fleischer for this reference. Suggestively, Taşköprüzāde (d. 1561) too classes the various historiographical sciences with the linguistic sciences, though not with any occult sciences, and certainly not under the rubric of applied mathematics, despite his clear reliance on Ibn Turka and al-Biṣṭāmī both – indeed, he is the first encyclopedist in the Arabo-Persian tradition to explicitly adopt Ibn Turka’s *tashkīk al-ḥarf* schema as a means of organizing all knowledge (Melvin-Koushki, “Of Islamic Grammatology,” pp. 89–91).

147 For a study of this work see Gardiner, “Lettrism and History.”

148 Biṣṭāmī’s startling move epitomizes a longer process peculiar to Persian scholarly culture, whereby encyclopedists, from Fakhr al-Dīn Rāzī (d. 1209) onward, increasingly shifted various occult sciences (particularly astrology, lettrism and geomancy) from the natural to the mathematical sciences in classifications of the sciences (sg. *taṣnīf al-‘ulūm*) – the surest index, I argue, for the renaissance of Pythagoreanism here in view. This process is traced in Melvin-Koushki, “Powers of One.”

149 Quinn, *Historical Writing*, pp. 46–53.

150 *Ibid.*, pp. 57–58. On his geomantic manual see Melvin-Koushki, “Persianate Geomancy,” p. 171.

of contemporary occult science, and featuring such eminent Safavid philosophers as Ghiyāṣ al-Dīn Maṣṣūṣ Dashtakī (d. 1542) and especially Shaykh Bahā'ī as talismanists in the service of empire – and explicitly connecting them to the New Brethren as inspired by Akhlāṭī.¹⁵¹

But as with architecture, the distinctively Timurid dual lettrist-astrological ideological platform is even more thoroughgoing in its Indo-Timurid (Mughal) iteration. Here again Shāhjahān's imperial patronage was pivotal. This, the grandson of the talismanic sovereign Akbar, commissioned a new star table, the *Zīj-i Shāhjahānī*, to celebrate his accession as Second Lord of Conjunction (*ṣāhib-qirān-i ṣānī*). Formally an updated and corrected version of Ulugh Beg's definitive *Zīj-i jadīd-i sulṭānī*, its introduction, for the first time in the history of the genre, makes explicit what is merely implicit in its fifteenth-century source: the proper scientific context for the mathematical study of the heavens – and hence dominion over the earth – is precisely lettrism.¹⁵²

Given the enduring status of Yazdī's *Book of Conquest* as model Persian imperial chronicle down to the colonial period, further research will doubtlessly turn up other instances long obscured by the reflexive occultophobia that still perverts the study of all things Islamic. And while in most cases later Persian (and Turkish and Arabic) historians' invocations of these occult sciences remained rather more eclectic and unsystematic than Yazdī's, we may nevertheless, I think, speak of the emergence of an explicit form of *Pythagorean historiography* – one peculiar to the early modern Persian cosmopolis, and unparalleled in the Latin or English, the labors of equally outspoken Pythagorean-occultists and imperial ideologue-historians like Dee all notwithstanding.

Imperial Reincarnationism and Magical Vegetarianism

He brings forth the living from the dead, and brings forth the dead from the living, and He revives the earth after it is dead; even so you shall be brought forth.

Quran 30:19

151 Subtelny, "Kāshifī's *Asrār-i qāsimī*," pp. 299–301. On Dashtakī's attempt to discredit his father's rival Davānī – an Akhlāṭīan-Ibn Turkīan lettrist – by astrological, lettrist and physiognomical argument see Bdaīwī, "Late Antique Intellectualism."

152 Moin, *The Millennial Sovereign*, pp. 23–24; Melvin-Koushki, "Timurid-Mughal Philosopher-Kings." To the same end, it proves the ontological equivalence of Temūr and Shāhjahān by means of lettrist analysis alone.

Similarly unparalleled in early modern Christendom is the resurgence of *reincarnationism* across the contemporary Persianate world, and especially Iran and India. The doctrine of reincarnation or metempsychosis (*tanāsukh*), of course, had venerable precedent in the heartlands of early Islam, being associated with Pythagoras and Plato on the one hand and the Sabeans of Harran on the other¹⁵³ – thus the Old Brethren’s unqualified embrace of the same.¹⁵⁴ While Ibn Turka and his fellow New Brethren did not likewise openly confess themselves to be reincarnationists, despite their similarly fervent Pythagorean commitments, I would suggest that their failure to *disavow* the doctrine, together with their robustly astrological-cyclicist theory of history and equally robustly transhumanist focus on theurgy, is itself telling vis-à-vis their early fifteenth-century context: for *takfirism* (religious denunciation as political strategy) was then rife, and even eminent Sunni scholar-jurists like Ibn Turka could be and were tarred with accusations of unorthodoxy, even Isma‘ilism, by jealous rivals with some success.¹⁵⁵

Indeed, the sharpest thorn in Ibn Turka’s side was precisely the Ḥurūfiyya, an anarchic Sufi group who took lettrism in a decidedly less scientific direction: its nativist founder, Faḡl Allāh Astarābādī (d. 1394), declared the beginning of a new age of divinity, wherein Persian now replaced Arabic as key to the Quran and the cosmos.¹⁵⁶ The New Brethren accordingly execrated the Ḥurūfīs for giving their master science a bad name, and identified themselves as Ḥarfīs (*ḥarfiyya*, *ahl al-ḥarf*) in contrast.¹⁵⁷ This fringe Sufi movement was eventually largely absorbed by the Bektāshīyya in Anatolia, and never attained much in the way of political influence; but its most important offshoot, the breakaway Nuṭṭavī or “Dottist” movement of Faḡl Allāh’s disciple Maḥmūd Paṣīkhānī (d. 1428), did.

Most importantly for our present purposes, moreover, Nuṭṭavī doctrine, unlike that of the New Brethren, was openly, even stridently, materialist-reincarnationist. At the same time, these Dottists too seem to have pursued an

153 Izdebska, “The Attitudes of Medieval Arabic Intellectuals”; on other varieties of early Irano-Islamic reincarnationism see Crone, *The Nativist Prophets*, ch. 12.

154 De Callatāy, *Ikhwan al-Safa’*, pp. 30–31. On the various Islamicate iterations of this doctrine generally see Asatryan, “Tanāsok,” *Elr*.

155 Reincarnationism came to be especially associated with Isma‘ili Shī‘ism from Ghazālī (d. 1111) onward (Walbridge, *Leaven of the Ancients*, p. 208) – hence the unconvincing attempt by Suhrawardī himself, father of Illuminationism, to distance himself from the doctrine (*idem*, *Wisdom of the Mystic East*, pp. 73–83).

156 Mir-Kasimov, *Words of Power*; Bashir, *Fazlallah Astarabadi*.

157 Fleischer, “Ancient Wisdom”; Melvin-Koushki, “The Quest,” ch. 8. *Ḥurūf* is the plural of Arabic *ḥarf*, ‘letter.’

imperialist project: evidence suggests that Shah ‘Abbās the Great (r. 1587–1629) flirted with Nuṭṭavism as a means of putting Safavid imperial ideology on a firmer and more competitive base, and perhaps even identified for a time as a Nuṭṭavī himself. But in the end he rejected their imperial platform as a threat to his personal supremacy, and conducted a brutal, hereticidal campaign against the movement. Thus spurned, the Nuṭṭavīs – along with the affiliated Āzarī movement of the Shirazi neo-Zoroastrian ideologue and occult scientist known pseudonymously as Āzar Kayvān (d. 1618), a sometime associate of Mir Dāmād and other leading Safavid philosophers – decamped for India, and in the Mughal emperor Akbar found a patron far more receptive.¹⁵⁸

Akbar, of course, is the most infamous of the great early Muslim modern sovereigns for his eager imperial embrace of sheerest heresy. His court ritual was entirely astral-magical, pivoting on planetary worship; his radically ecumenical doctrine of Universal Harmony (*ṣulḥ-i kull*) – an alchemical-medical reference¹⁵⁹ – demoted Islam to legal and sociopolitical parity with Hinduism, Jainism and other religions of the empire; and he instituted the Divine Religion (*dīn-i ilāhī*), a reincarnationist, vegetarianist court cult centered on the worship of his person as divine.¹⁶⁰ Now specialist historians long facetiously assumed such shocking innovations to be a mere product of Hindu-Islamic syncretism; but new research has shown them to be inspired in the main precisely by such radically *Persocentric* occultist movements as the Nuṭṭavīs and Āzarīs, both born in Timurid-Safavid Iran, not India.¹⁶¹ Akbar’s “heresy” in general and reincarnationism in particular therefore cannot be read as a rejection of Islam and Iran, but rather as an especially efficient (because bisemous) means of signaling his performace of a specifically Timurid form of *Islamic-Iranian talismanic kingship* – after the philhellenic lettrist-astrological model first formulated by Ibn Turka and his fellow New Brethren.¹⁶²

And they too were *vegetarianists*: Akhlāṭī, Ibn Turka and (to my knowledge) all subsequent Persian grimoire authors are emphatic about the need to maintain a vegetarian diet (*tark-i akl-i ḥayavānī*) to ensure the effectiveness of

158 On the Nuṭṭavīs see e.g. Babayan, *Mystics, Monarchs, and Messiahs*, passim; on the Āzarīs see Sheffield, “Lord of the Planetary Court.”

159 My thanks to Daniel Sheffield for this observation. See now the special issue of *Modern Asian Studies* (56, no. 3 [2022]) on this theme edited by Azfar Moin, and particularly Sheffield’s contribution, “Exercises in Peace.”

160 On this cult see Moin, *The Millennial Sovereign*, ch. 5.

161 Sheffield, “Lord of the Planetary Court.”

162 Melvin-Koushki, “Early Modern Islamicate Empire.”

letter-magical and letter-divinatory operations.¹⁶³ Given the present state of research, we do not know when this doctrine first became *sine qua non* for the practice of Persian magic, or the extent to which it might be considered evidence for Indian influence, rather than a simple continuation of Western precedent.¹⁶⁴ But as all good early modern perennialists knew, Muslim or Christian, the vegetarian mage Pythagoras himself was the disciple of Zarathustra/Zoroaster, Iranian magus supreme and source of all ancient wisdom.¹⁶⁵ What we can say for certain, then, is that the New Brethren of Purity influentially pushed lettrism, vegetarianism and perhaps even reincarnationism not as crypto-Hindus, but as faithful – albeit presumably bean-eating – Persianate Muslim disciples of the Greek-*cum*-Persian sage Pythagoras.

Toward a New Pythagoreanism?

It from bit.

JOHN WHEELER, “Information, Physics, Quantum”

The entire universe ... is perfused with signs, if it is not composed exclusively of signs.

C. S. PEIRCE, “The Basis of Pragmatism”

Magic no longer works, if it ever did, or so we are told; yet we are now living in a veritable golden age of magic.¹⁶⁶ By the same token, the modern era would seem to be a golden age of Pythagoreanism. Cosmological arguments for the world as a linguistic-mathematical text – and hence subject to decoding and (magical) recoding – are again flourishing in Euro-American elite and popular culture alike. They have likewise persisted in the Islamic world in the teeth of colonialism, reformism, nationalism and salafism, albeit in pockets or

163 A full list of ascetic requirements is given by Kāshifī's son and fellow author on lettrism, 'Alī Šafī (d. 1533), who was largely responsible for the Safavid reception of this Timurid science, in his *Tuhfa-yi Majdīyya*; see Melvin-Koushki, “Qizilbash Magic.”

164 It is here suggestive that Mīr Dāmād's close colleague and fellow philosopher Mīr Findiriskī (d. 1640), who pursued his interests in alchemy and yoga to India and became a Sanskrit-Persian translator as such, seems to have adopted vegetarianism more as an Indic practice than as a Pythagorean one; my thanks to Mathieu Terrier for this observation.

165 My thanks to Daniel Sheffield for this observation; see his *Cosmopolitan Zarathustras*.

166 A less offensive term is *psychophysics*. See Josephson-Storm, *The Myth of Disenchantment*; Radin, *The Science of Magic*; idem, *Real Magic*.

underground – until the Internet revolution: fittingly, the Islamic Occult now everywhere manifests in zeroes and ones.¹⁶⁷

A number of recent developments are here suggestive. Not surprisingly, the epicenter of this new, suspiciously Pythagorean turn is the discipline of physics, which has long since displaced lettrist-kabbalist metaphysics as queen of the sciences – due, ironically, to its mathematization by the Pythagorean occultist Newton, who would surely have appreciated the sheer weirdness of quantum mechanics, for all that it has overturned his neat cosmology. And now the last decades have seen the upsurge of a new *physics of information*.¹⁶⁸ This ontogrammatological turn is epitomized by Princeton physicist John Wheeler's famous 1989 dictum *It from bit* – that is to say, “all things physical are information-theoretic in origin and this is a *participatory universe*.”¹⁶⁹ Most shockingly, this emergent cosmology requires us to recognize the universe as a “metareality of information structures,” and the unidirectional flow of time and the strict limits of space as human constructs; hence the ability of human consciousness, *logos* processor that it is, to quantum-mechanically change physical reality by the mere act of observation, even in the past.¹⁷⁰ Evolutionary theologians have seized upon this new physics of information in turn as the only workable means of reconciling the Christian doctrine of creation with Darwinian evolution (shades of Pico's embrace of kabbalah in pursuit of a project equally paradoxical): the universe as meaning-generating device.¹⁷¹ Likewise, Peircean panpsychist semiotics – wherein every existent is a sign, and parapsychology a necessary line of scientific inquiry – has become a cottage industry in philosophy.¹⁷² More, some cognitive scientists have mathematically hypothesized an almost idealist *conscious realism*, whereby perception alone erects time and space and quantum-mechanically codes

167 Doostdar, *The Iranian Metaphysicals*; Taneja, *Jinnealogy*. A classic example is that of the New Brother Maḥmūd Dihdār himself, whose magnum opus *Keys to All Locks* (*Mafātīḥ al-maghālīq*), a seminal lettrist grimoire, has become a bestseller online to an extent the author could never have dreamed.

168 See e.g. Vedral, *Decoding Reality*.

169 Wheeler, “Information, Physics, Quantum,” p. 5 (*italics original*).

170 Vallée, “A Theory of Everything (Else).”

171 For examples see Davies and Gregersen (eds), *Information and the Nature of Reality*; Haught, *The New Cosmic Story*.

172 See e.g. Peirce's “The Basis of Pragmatism.” On contemporary panpsychist philosophy generally see e.g. Goff, *Galileo's Error*; on the pressing need for modern historians of occultism to take more seriously the usually panpsychist cosmology of their historical actors see Melvin-Koushki, “Is (Islamic) Occult Science Science?”

what we take to be physical reality.¹⁷³ And geneticists persist in speaking of the Book of Life in suspiciously kabbalist fashion¹⁷⁴ – indeed, the former director of the Human Genome Project at the NIH, Francis Collins, is an avowed Christian kabbalist.¹⁷⁵ As for the transhumanist mages and alchemists – we call them technocrats – of Silicon Valley (again with the geomantic sand!), who too prefer practice to theory, they labor toward the digital Singularity and diligently plan their ascent to the celestial sphere of Mars.

But the atomized, balkanized and ghettoized nature of modern academic culture, wherein gestures toward interdisciplinarity merely mask the all-structuring rictus of strict disciplinarity, militates against the acknowledgment and analysis of such developments as a coherent cultural current: unlike their early modern forebears, contemporary humanists, theologians and scientists rarely have in common the language of *mathematics*, and certainly usually do not openly identify as Pythagorean. As such, they are rendered illegible to the modern historian of Pythagoreanism.

No such debility affects popular or literary culture, however, where imagination and intuition are less starkly bounded; and it would seem to be consciously structured by the same Neo-neopythagorean ethos to a similar extent. Here the film *The Matrix* (1999) is a classic, Neognostic case in point, as is the more recent *Doctor Strange* (2016), which features historically semi-accurate talismanic operations as essential to theosis. More programmatically, influential comic book author and ritual mage Alan Moore's *Promethea* (2005) gorgeously pushes an explicitly kabbalist-alchemical project as solution to the ills of Euro-American modernity, which has yet to add *coagula* to *solve* and so complete the Great Work.¹⁷⁶ As for high modern literary culture: its apotheosis, James Joyce's *Finnegans Wake*, has been shown to have a perfectly mathematically multifractal structure – which is to say, to mimic the mathematical patterns governing most natural phenomena. It appears this miracle of mimesis was very consciously achieved, moreover, in that Joyce declared his masterpiece to have finally realized the ancient, impossible Pythagorean dream of squaring the circle.¹⁷⁷

All of which sounds remarkably *lettrist*; Ibn Turka would have grounds to be smug. This is not to suggest, however, that such instances of modern New

173 Thus Donald D. Hoffman, cognitive scientist at the University of California, Irvine, and author of *Visual Intelligence*.

174 Von Stuckrad, "Rewriting the Book of Nature."

175 See e.g. Collins's *The Language of God*.

176 Thus Moore's alchemical-kabbalist critique of scientific modernity in *Promethea* 32 (April 2005).

177 Drozd et al., "Quantifying Origin." For a Joycean reading of Iskandar's horoscope see Melvin-Koushki, "Selenocentrism and Heliocentrism."

Pythagoreanism do not also radically diverge from early modern and ancient Pythagorean legacies alike. They are now rarely couched, for example, in specifically Neoplatonic terms, and tend not to feature reincarnationism, vegetarianism or a horror of beans. And even that most kabbalist of recent scholarly trends, Derridean deconstruction, which has forced its own seachange in a host of disciplines in the humanities (the linguistic turn), disavows the lettrist ethical-technological imperative to *reconstruct the world* too.¹⁷⁸

As that may be, the disciples of Pythagoras, whether consciously or unconsciously so, continue to abound at the heart of modern Western culture. The Copernican Revolution was a fundamentally Pythagorean project, as intellectual historians have shown; how will historians centuries hence explain our Information Revolution, much less AI? (The geomancer's quest for ACI, Artificial Cosmic Intelligence, obviously rhymes with the current quest for AGI – and very much put it to the same technocratic and imperial uses.) For to be Western is to philosophize in simultaneously mathematical and linguistic terms;¹⁷⁹ and we Moderns are in hellbent Starstraining pursuit of mathematically-linguistically driven *transcendence* just as much as our Ancients.

This emphatically includes Islam as historically the largest vehicle for Western, Hellenic-Abrahamic cultures, exemplified by the New Brethren and their direct and indirect heirs to the present, for all that these have been totally ignored by historians of Pythagoreanism to date. I have here made a case for why this must change, why we must not persist in writing the greater part of Pythagorean thought out of Western history. To do so is to fall prey to the colonialist-Orientalist deadfall – ironically of European Renaissance anti-Ottoman vintage – whereby Islam can never be the West, and Muslims can never truly be Greek, much less Modern – despite their constant declarations to the contrary.¹⁸⁰

The epistemicidal effectiveness of that deadfall has meant that we cannot yet trace the many evolutions of Islamic Pythagoreanism in the so-called post-classical period (a polemical designation) generally, nor the legacy of the New Brethren specifically, the most influential Pythagorean-imperialists of the early modern Greater West. But the massive surviving archive nevertheless makes it clear that their legacy continued hale and hearty down to and through the epistemic and sociopolitical ruptures that were European colonialism and Protestantizing Islamic reformism – as does the fact that

178 Melvin-Koushki, "Of Islamic Grammatology," pp. 43–44, 95–97; Hanegraaff, *Hermetic Spirituality*, passim.

179 Melvin-Koushki, "*Tahqīq* vs. *Taqlīd*," pp. 235–36.

180 Ibid., passim.

reformists were sometimes lettrists too.¹⁸¹ Indeed, lettrism itself – in a specifically Timurid imperial mode – was invoked in the late nineteenth-century Persianate world as a primary means of anticolonial resistance; and geomancy too has handily survived the capitalization and now Black-Mirrorization of the world.¹⁸²

The intervening seventeenth and eighteenth centuries were difficult ones, to be sure, when the Persian cosmopolis was wracked anew by plague and inflation. High Pythagorean occult philosophy accordingly waned, as its scholarly exponents lost the leisure to pursue it; but occult-technological practice certainly did not, at least in certain quarters, and despite a lack of new theoretical developments in the Pythagorean sciences.¹⁸³ And the nineteenth century saw even theory waxing again, which gave rise in turn to the Shaykhiyya movement, and thence Bābism and the now global Bahā'ī Faith, whose scriptures and prayer technologies too are thoroughlygoingly lettrist.¹⁸⁴

As for Ibn Turka: his status as preeminent early modern authority on Pythagorean science remained intact in Iran through at least the high Qajar period, when Mullā 'Alī Nūrī (d. 1830), that great reviver of Ṣadrian and Mīr Dāmādiān philosophy, avidly glossed Ibn Turka's *Investigations* – the first Western manifesto to successfully call for the mathematization of the cosmos.¹⁸⁵ Predictably, the fullthroatedly imperial tenor of Doctor Littleturk's Pythagoreanism was lost to his Muslim heirs during the age of Christian and Soviet colonialism. Yet they lost it precisely to such upstart entities as the British Empire, among others – itself the long-delayed realization of the bihemispheric imperial vision of Doctor Dee, that Welsh Ibn Turka.

Kepler, in short, was far from the last Pythagorean.¹⁸⁶

181 No less a reformist than Shāh Walī Allāh of Delhi (d. 1762), putative father of the Deoband movement, embraced lettrism specifically and the occult sciences generally as central to his life's project; see Morgan, "Spokesman of the Unseen World." On broader Indian Ocean developments see AlMaazmi, "An Enchanted Sea."

182 Melvin-Koushki and Pickett, "Mobilizing Magic"; Melvin-Koushki and Cummins, "Geomancing Is Googling."

183 See e.g. Küçük, *Science without Leisure*; for a case in point see Melvin-Koushki, "Persianate Geomancy." On the gradual loss in the music of Persianate India of the astral-magical component of *raga* see Schofield, "Music, Art and Power."

184 Cole, "The World as Text"; Melvin-Koushki, "The Quest," 281; idem, "Shaykhi Prayer." Following Safavid precedent, they embraced alchemy and lettrism as an integral Imamic and scientific pair; see Melvin-Koushki, "Powers of One," pp. 167–68.

185 On Nūrī's firm commitment to Ibn Turkian lettrism see Rizvi, "Mullā 'Alī Nūrī."

186 Thus Field, *Kepler's Geometrical Cosmology*, 170. Cf. Carl Huffman, "Pythagoreanism," *SEP*, §5.2, who allows that "many modern scientists accept the basic tenet that knowledge of the natural world is to be expressed in mathematical formulae, which is rightly

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Bibliography

- Acevedo, Juan, *Alphanumeric Cosmology from Greek into Arabic: The Ideo of Stoicheia through the Medieval Mediterranean*, Tübingen: Mohr Siebeck, 2020.
- Adams, Peter Mark. *Hagia Sophia / Sanctum of Kronos: Spiritual Dissent in an Age of Tyranny*. London: Scarlet Imprint, 2023.
- Agrippa, Heinrich Cornelius, *De occulta philosophia libri tres*, ed. Vittoria Perrone Compagni, Leiden: Brill, 1992.
- Ahmed, Shahab, *What Is Islam? The Importance of Being Islamic*, Princeton: Princeton University Press, 2017.
- Albertson, David, *Mathematical Theologies: Nicholas of Cusa and the Legacy of Thierry of Chartres*, Oxford: Oxford University Press, 2014.
- Albertson, David (ed.), *Cusanus Today: Thinking with Nicholas of Cusa Between Philosophy and Theology*. Washington, DC: The Catholic University of America Press, 2024.
- AlMaazmi, Ahmed Y. "An Enchanted Sea: Occultism, Empire and Society in the Western Indian Ocean, 1450–1750", PhD dissertation, Princeton University, 2025.
- Asatryan, Mushegh, "Tanāsok," *Encyclopaedia Iranica Online*.
- Babayan, Kathryn, *Mystics, Monarchs, and Messiahs: Cultural Landscapes of Early Modern Iran*, Cambridge, Mass.: Harvard University Press, 2002.
- Bacon, Francis, *A Harmony of the Essays*, ed. Edward Arber, London: Constable, 1871.
- Bacon, Francis, *New Atlantis*, ed. William Rawley, London: n.p., 1627.
- Bahār, Muḥammad-Taqī, *Sabk-shināsī, yā tārikh-i taṭavvur-i naṣr-i fārsī*, 3 vols, Tehran: Amīr Kabīr, 1381 Sh./2002.
- Balivet, Michel, *Islam mystique et révolution armée dans les Balkans Ottomans: vie du Cheikh Bedreddīn le "Hallāj des Turcs" (1358/59–1416)*, Istanbul: Isis, 1995.
- Bashir, Shahzad, *Fazlallah Astarabadi and the Hurufis*, Oxford: Oneworld, 2005.
- Bauer, Thomas, *Die Kultur der Ambiguität: Eine andere Geschichte des Islams*, Berlin: Verlag der Weltreligionen, 2011.
- Bauer, Thomas, "Mamluk Literature: Misunderstandings and New Approaches," *Mamlūk Studies Review* 9/2 (2005), pp. 105–32.

regarded as a central Pythagorean thesis ... and may, in a rudimentary form, go back to Pythagoras himself."

- Bdaiwi, Ahab, "Late Antique Intellectualism in Medieval Islam: The Shiraz Circle and the Revival of Ancient and Islamic Knowledge," *Global Intellectual History* 8 (2023), forthcoming.
- Begley, W. E. and Z. A. Desai, *Taj Mahal: The Illumined Tomb: An Anthology of Seventeenth-Century Mughal and European Documentary Sources*, Cambridge, Mass.: The Agha Khan Program for Islamic Architecture, 1989.
- Binbaş, İ. Evrim, *Intellectual Networks in Timurid Iran: Sharaf al-Dīn 'Alī Yazdī and the Islamic Republic of Letters*, Cambridge: Cambridge University Press, 2016.
- Bloom, Jonathan M., review of *The Timurid Architecture of Iran and Turan*, by Lisa Golombek and Donald Wilber, *The Journal of the Society of Architectural Historians* 48/3 (1989), pp. 303–4.
- Borrut, Antoine, *Astrology and History in Early Islam: Aligning Heaven and Earth*, Edinburgh: Edinburgh University Press, 2026.
- Caiazzo, Irene, Constantinos Macris and Aurélien Robert (eds), *Brill's Companion to the Reception of Pythagoras and Pythagoreanism in the Middle Ages and the Renaissance*, Leiden: Brill, 2022.
- Caiozzo, Anna, "The Horoscope of Iskandar Sultān as a Cosmological Vision in the Islamic World," *Horoscopes and Public Spheres: Essays on the History of Astrology*, ed. Günther Oestmann, H. Darrel Rutkin and Kocku von Stuckrad, Berlin: de Gruyter, 2005, pp. 115–44.
- Casewit, Yousef, *The Mystics of al-Andalus: Ibn Barrajān and Islamic Thought in the Twelfth Century*, Cambridge: Cambridge University Press, 2017.
- Collins, Francis S., *The Language of God: A Scientist Presents Evidence for Belief*, New York: Free Press, 2006.
- Chann, Naindeep, "Lord of the Auspicious Conjunction: Origins of the *Ṣāhib-Qirān*," *Iran and the Caucasus* 13 (2009), pp. 93–110.
- Cole, Juan, "The World as Text: Cosmologies of Shaykh Ahmad al-Ahsa'i," *Studia Islamica* 80 (1994), pp. 145–63.
- Cornelli, Gabriele, *In Search of Pythagoreanism: Pythagoreanism as an Historiographical Category*, Berlin: De Gruyter, 2013.
- Crone, Patricia, *The Nativist Prophets of Early Islamic Iran: Rural Revolt and Local Zoroastrianism*, Cambridge: Cambridge University Press, 2014.
- Dānīshpazhūh, Muḥammad-Taqī, "Majmū'a-yi rasā'il-i Khujandī," *Farhang-i Īrān-zamīn* 14 (1345–46 Sh./1966–67), pp. 307–12.
- Davies, Paul and Niels Henrik Gregersen (eds), *Information and the Nature of Reality: From Physics to Metaphysics*, Cambridge: Cambridge University Press, 2010.
- de Callatāy, Godefroid, "Rutba vs Ġāya: A Reconsideration of the Impact of the *Rasā'il Iḥwān al-Ṣafā'* on Maslama b. Qāsim al-Qurṭubī," *Arabica* 68 (2021), pp. 495–509.
- de Callatāy, Godefroid, "Magia en al-Andalus: *Rasā'il Iḥwān al-Ṣafā'*, *Rutbat al-ḥakīm y Gāyat al-ḥakīm* (Picatrix)," *Al-Qanṭara* 34/2 (2013), pp. 297–344.

- de Callataj, Godefroid, *Ikhwan al-Safa': A Brotherhood of Idealists on the Fringe of Orthodox Islam*, Oxford: Oneworld, 2005.
- Dee, John, *Monas Hieroglyphica*, trans. C. H. Josten, *Ambix* 12 (1964), pp. 84–221.
- Dee, John, *Mathematicall Praeface*, London: n.p., 1570.
- De León-Jones, Karen S., *Giordano Bruno and the Kabbalah: Prophets, Magicians and Rabbis*, New Haven: Yale University Press, 1997.
- De Smet, Daniel, “Les philosophes grecs, tous monothéistes! Une relecture néoplatonicienne islamisée de l’histoire de la philosophie (Pseudo-Ammonius),” *Appropriations monothéistes de figures “païennes,”* eds C. Macris, J. Soler and A. Van den Kerchove, special issue of *Revue de l’histoire des religions* 236/4 (2019), pp. 821–46.
- Domnitch, Evelina and Dmitri Gelfand, “Hyperspectral Percipience,” *Digimag Journal* 74 (Winter 2013), pp. 76–84.
- Doostdar, Alireza, *The Iranian Metaphysicals: Explorations in Science, Islam, and the Uncanny*, Princeton: Princeton University Press, 2018.
- Drozdz, Stanislaw *et al.*, “Quantifying Origin and Character of Long-Range Correlations in Narrative Texts,” *Information Sciences* 331 (February 2016), pp. 32–44.
- Ebstein, Michael, *Mysticism and Philosophy in al-Andalus: Ibn Masarra, Ibn al-‘Arabī and the Ismā‘īlī Tradition*, Leiden: Brill, 2014.
- Eglash, Ron, *African Fractals: Modern Computing and Indigenous Design*, New Brunswick: Rutgers University Press, 1999.
- Elmore, Gerald T., *Islamic Sainthood in the Fullness of Time: Ibn al-‘Arabī’s Book of the Fabulous Gryphon*, Leiden: Brill, 1999.
- Emami, Farshid, *Isfahan: Architecture and Urban Experience in Early Modern Iran*, University Park: The Pennsylvania State University Press, 2024.
- Emami, Farshid, “Religious Architecture of Safavid Iran,” *The Religious Architecture of Islam*, vol. 1, eds Hasan-Uddin Khan and Kathryn Blair Moore, Turnhout: Brepols, 2021, pp. 257–73.
- Emami, Farshid, “Discursive Images and Urban Itineraries: Literary Form and City Experience in Early Modern Iran,” *Journal for Early Modern Cultural Studies* 18/3 (2018), pp. 154–85.
- Endress, Gerhard, “Mathematics and Philosophy in Medieval Islam,” *The Enterprise of Science in Islam: New Perspectives*, eds J. P. Hogendijk and A. I. Sabra, Cambridge, Mass.: MIT Press, 2003, pp. 121–27.
- Eryılmaz, Sinem, “Following the Steps of the *Ikhwān al-Ṣafā* in the Ottoman World III: Insights from Lamii Çelebi’s Works,” forthcoming.
- Esots, Janis, *Patterns of Wisdom in Safavid Iran: The Philosophical School of Isfahan and the Gnostic of Shiraz*, London: I.B. Tauris in association with the Institute of Ismaili Studies, 2021.
- Fazlıoğlu, İhsan, “Between Reality and Mentality: Fifteenth Century Mathematics and Natural Philosophy Reconsidered,” *Nazariyat* 1 (2014), pp. 1–39.

- Fazlhoğlu, İhsan, "The Samarqand Mathematical-Astronomical School," *Journal for the History of Arabic Science* 14 (2008), pp. 3–68.
- Fazlhoğlu, İhsan, "İlk Dönem Osmanlı İlim ve Kültür Hayatında İhvanu's-Safâ ve Abdurrahman Bistâmî," *Dîvân: İlmî Araştırmalar* 1 (1996), pp. 229–40.
- Field, J. V., *Kepler's Geometrical Cosmology*, London: The Athlone Press, 1988.
- Fierro, Maribel, "Bâtînism in al-Andalus: Maslama b. Qāsim al-Qurṭubî (d. 353/964), Author of the *Rutbat al-Ḥakīm* and the *Ghāyat al-Ḥakīm* (*Picatrix*)," *Studia Islamica* 84/2 (1996), pp. 87–112.
- Fleischer, Cornell H., "A Mediterranean Apocalypse: Prophecies of Empire in the Fifteenth and Sixteenth Centuries," *Speaking the End Times: Prophecy and Messianism in Early-Modern Eurasia*, ed. Mayte Green-Mercado, special double issue of *Journal of the Economic and Social History of the Orient* 61/1–2 (2018), pp. 18–90.
- Fleischer, Cornell H., "Ancient Wisdom and New Sciences: Prophecies at the Ottoman Court in the Fifteenth and Early Sixteenth Centuries," *Falnama: The Book of Omens*, eds Massumeh Farhad and Serpil Bağcı, London: Thames & Hudson, 2009, pp. 232–43.
- Fleischer, Cornell H., "Shadow of Shadows: Prophecy in Politics in 1530s Istanbul," *International Journal of Turkish Studies* 13 (2007), pp. 51–62.
- Fleischer, Cornell H., "Royal Authority, Dynastic Cyclism, and 'Ibn Khaldûnism' in Sixteenth-Century Ottoman Letters," *Journal of Asian and African Studies* 18/3–4 (1983), pp. 198–220.
- Fowden, Garth, *Before and after Muḥammad: The First Millennium Refocused*, Princeton: Princeton University Press, 2013.
- Gardiner, Noah, "Lettrism and History in 'Abd al-Raḥmān al-Bisṭāmī's *Naẓm al-sulūk fī musāmārat al-mulūk*," *Islamicate Occult Sciences in Theory and Practice*, eds Liana Saif, Francesca Leoni, Matthew Melvin-Koushki and Farouk Yahya, Leiden: Brill, 2021, pp. 230–66.
- Gardiner, Noah, *Ibn Khaldūn versus the Occultists at Barqūq's Court: The Critique of Lettrism in al-Muqaddimah*, Berlin: EB-Verlag, 2020.
- Gardiner, Noah, "The Occultist Encyclopedism of 'Abd al-Raḥmān al-Bisṭāmī," *Mamlūk Studies Review* 20 (2017), pp. 3–38.
- Gardiner, Noah, "Esotericism in a Manuscript Culture: Aḥmad al-Būnī and His Readers through the Mamlūk Period," PhD diss., University of Michigan, 2014.
- Gharipour, Mohammad, "The Gardens of Safavid Isfahan and Renaissance Italy: A New Urban Landscape?" *Gardens of Renaissance Europe and the Islamic Empires: Encounters and Influences*, ed. Mohammad Gharipour, University Park: Pennsylvania State University Press, 2017, pp. 101–34.
- Goff, Philip, *Galileo's Error: Foundations for a New Science of Consciousness*, London: Rider, 2019.
- Greer, John Michael, *The Art and Practice of Geomancy: Divination, Magic, and Earth Wisdom of the Renaissance*, San Francisco: Weiser, 2009.

- Gutas, Dimitri, *Greek Thought, Arabic Culture: The Graeco-Arabic Translation Movement in Baghdad and Early 'Abbāsīd Society (2nd–4th/8th–10th Centuries)*, New York: Routledge, 1998.
- Håkansson, Håkan, *Seeing the Word: John Dee and Renaissance Occultism*, Lund: Lunds Universitet, 2001.
- Hallyn, Fernand, *The Poetic Structure of the World: Copernicus and Kepler*, trans. Donald M. Leslie, New York: Zone Books, 1990.
- Hanegraaff, Wouter J., *Esotericism in Western Culture: Counter-Normativity and Rejected Knowledge*, London: Bloomsbury, 2025.
- Hanegraaff, Wouter J., *Hermetic Spirituality and the Historical Imagination: Altered States of Knowledge in Late Antiquity*, Cambridge: Cambridge University Press, 2022.
- Hanegraaff, Wouter J., *Esotericism and the Academy: Rejected Knowledge in Western Culture*, Cambridge: Cambridge University Press, 2012.
- Hanegraaff, Wouter J., "Beyond the Yates Paradigm: The Study of Western Esotericism between Counterculture and New Complexity," *Aries* 1/1 (2001), pp. 5–37.
- Harris, Nicholas, "Better Religion through Chemistry: Aydemir al-Jildakī and Alchemy under the Mamluks," PhD diss., University of Pennsylvania, forthcoming 2026.
- Haight, John F., *The New Cosmic Story: Inside Our Awakening Universe*, New Haven: Yale University Press, 2017.
- Heninger Jr., S. K., *Touches of Sweet Harmony: Pythagorean Cosmology and Renaissance Poetics*, San Marino, Cal.: The Huntington Library, 1974.
- Henry, John, *Knowledge Is Power: How Magic, the Government and an Apocalyptic Vision Inspired Francis Bacon to Create Modern Science*, Lanham, MD: Icon Books, 2003.
- Hersey, G. L., *Pythagorean Palaces: Magic and Architecture in the Italian Renaissance*, Ithaca: Cornell University Press, 1976.
- Hoffman, Donald D., *Visual Intelligence: How We Create What We See*, New York: Norton, 1998.
- Huffmann, Carl (ed.), *A History of Pythagoreanism*, Cambridge: Cambridge University Press, 2014.
- Huffmann, Carl, *Archytas of Tarentum: Pythagorean, Philosopher and Mathematician King*, Cambridge: Cambridge University Press, 2010.
- Huffmann, Carl, "Pythagoreanism," *Stanford Encyclopedia of Philosophy*, <https://plato.stanford.edu/en-tries/pythagoreanism>.
- Ikhwān al-Ṣafā', *Rasā'il*, 4 vols, Beirut: al-A'lamī, 2005.
- Irwin, Robert, "Al-Maqrīzī and Ibn Khaldūn, Historians of the Unseen," *Mamlūk Studies Review* 7/2 (2003), pp. 217–30.
- Iskandar Sulṭān, "Dībācha-yi Jāmi' al-sulṭānī," in Sharaf al-Dīn 'Alī Yazdī, *Munsha'āt*, eds Īraj Afshār and Muḥammad Riẓā Abū'ī Mahrīzī, Tehran: Ṣurayyā, 1388 Sh./2009, pp. 207–11.
- Izdebska, Anna, "The Attitudes of Medieval Arabic Intellectuals towards Pythagorean Philosophy: Different Approaches and Ways of Influence," *Cultures in Motion*:

- Studies in the Medieval and Early Modern Periods*, eds Adam Izdebski and Damian Jasiński, Cracow: Jagiellonian University Press, 2014, 25–44.
- Joost-Gaugier, Christiane, *Pythagoras and Renaissance Europe: Finding Heaven*, Cambridge: Cambridge University Press, 2014.
- Joost-Gaugier, Christiane, *Measuring Heaven: Pythagoras and His Influence on Thought and Art in Antiquity and the Middle Ages*, Ithaca: Cornell University Press, 2006.
- Josephson-Storm, Jason Ā., *The Myth of Disenchantment: Magic, Modernity, and the Birth of the Human Sciences*, Chicago: University of Chicago Press, 2017.
- Josten, C. H., “A Translation of John Dee’s ‘Monas Hieroglyphica’ (Antwerp, 1564), with an Introduction and Annotations,” *Ambix* 12 (1964), pp. 84–221.
- Kahn, Charles H., *Pythagoras and the Pythagoreans: A Brief History*, Indianapolis: Hackett, 2001.
- Kalvesmaki, Joel, *The Theology of Arithmetic: Number Symbolism in Platonism and Early Christianity*, Cambridge, Mass.: Center for Hellenic Studies/Harvard University Press, 2013.
- Kāzīrūnī, Abū l-Qāsim Anṣārī, *Sullam al-samāwāt*, ed. ‘Abd Allāh Nūrānī, Tehran: Mīrās-i Maktūb, 1386 Sh./2007.
- Killeen, Kevin and Peter J. Forshaw (eds), *The Word and the World: Biblical Exegesis and Early Modern Science*, Houndsmill, Basingstoke: Palgrave Macmillan, 2007.
- Koch, Ebba, *The Planetary King: Humayun Padshah, Inventor and Visionary on the Mughal Throne*, Ahmedabad: Mapin, 2022.
- Koch, Ebba, “Carved Pools, Rock-Cut Elephants, Inscriptions, and Tree Columns: Mughal Landscape Art as Imperial Expression and Its Analogies to the Renaissance Garden,” *Gardens of Renaissance Europe and the Islamic Empires: Encounters and Influences*, ed. Mohammad Gharipour, University Park: Pennsylvania State University Press, 2017, pp. 185–212.
- Koch, Ebba, *The Complete Taj Mahal and the Riverfront Gardens of Agra*, London: Thames & Hudson, 2006.
- Koch, Ebba, “The Taj Mahal: Architecture, Symbolism, and Urban Significance,” *Muqarnas* 22 (2005), pp. 128–49.
- Küçük, Harun, *Science without Leisure: Practical Naturalism in Istanbul, 1660–1732*, Pittsburgh: University Press, 2019.
- Losensky, Paul, *Welcoming Fighānī: Imitation and Poetic Individuality in the Safavid-Mughal Ghazal*, Costa Mesa: Mazda, 1998.
- Lu, Peter J. and Paul J. Steinhardt, “Decagonal and Quasi-Crystalline Tilings in Medieval Islamic Architecture,” *Science* 315, no. 5815 (2007), pp. 110–610.
- Markiewicz, Christopher, *The Crisis of Kingship in Late Medieval Islam: Persian Emigres and the Making of Ottoman Sovereignty*, Cambridge: Cambridge University Press, 2019.
- Markiewicz, Christopher, “History as Science: The Fifteenth-Century Debate in Arabic and Persian,” *Journal of Early Modern History* 21 (2017), pp. 216–40.

- Markiewicz, Christopher, "The Crisis of Rule in Late Medieval Islam: A Study of Idrīs Bidlīsī (861–926/1457–1520) and Kingship at the Turn of the Sixteenth Century," PhD diss., University of Chicago, 2015.
- Melvin-Koushki, Matthew, *Persian Pythagoreanism, Imperial Occultism and Mathematical-Linguistic Science in the Timurid Renaissance: The Lettrist Treatises of Ibn Turka*, Leiden: Brill, forthcoming 2026.
- Melvin-Koushki, Matthew, "Ibn Turka's (d. 835/1432) Pythagorean Elevation of Sight over Hearing as Vehicle of Total Perception," in *Islamic Sensory History*, vol. 3, eds Christian Lange and Yusuf Unal, Leiden: Brill, forthcoming.
- Melvin-Koushki, Matthew, "Selenocentrism and Heliocentrism in Early Modern Persianate Imperial Cultures: 'Alī *versus* Jesus, with Hermes Presiding. A Harranian Essay in Astro-Decoloniality," *Al-'Uṣūr al-Wuṣṭā* 33 (2025), forthcoming.
- Melvin-Koushki, Matthew, "Mir Dāmād's *On Doffing*: Safavid Philosophy, Ritual Magic and Out-of-Body Experience," *De la lettre à l'esprit / From the Letter to the Spirit: Studies in Honour of Mohammad Ali Amir-Moezzi*, eds Orkhan Mir-Kasimov and Mathieu Terrier, Turnhout: Brepols, 2025, vol. 2, pp. 779–818.
- Melvin-Koushki, Matthew, "The Occult Court: 'Alī Ṣafī's *Boon for the Khan* (1522) as Timurid-Safavid Manual of Boozing and Battling Magic and Marker of Early Modernity," *The Religious Cultures of Premodern Islamicate Courts in Transregional Perspective*, ed. Christian Mauder, special issue, *Al-Masāq* 37/1 (2025), forthcoming.
- Melvin-Koushki, Matthew, "Definition as (De)colonial Weapon: Western Esotericism Meets Islamic Occultism and Is Weirder Out," *Aries: Journal for the Study of Western Esotericism* 24 (2024), pp. 231–35.
- Melvin-Koushki, Matthew, "Lettrism, Astrology and Jazz Philosophy in Ibn 'Arabī's *Allusions of the Quran in the World of Man*," *Journal of the Muhyiddin Ibn 'Arabi Society* 73 (2023), pp. 53–75.
- Melvin-Koushki, Matthew, "Better than Sufi Sex: Ibn Turka on the Superiority of Lettrism to Sufism as Model of Occult Islamic Humanism," *Islam ed esoterismo/ Islam and Esotericism*, eds Michele Olzi and Lisa Pizzighella, special issue of *La Rosa di Paracelso* 2 (2020 [2022]), pp. 53–80.
- Melvin-Koushki, Matthew, "Occult Ecumenism: Maḥmūd Dihdār Shīrāzī's *Unveiling Secrets* as Exemplar of Timurid-Safavid Sunni-Shi'i Science," *Iranian Studies* 55/5–6 (2022), pp. 1–31.
- Melvin-Koushki, Matthew, "Translating Esotericism: Early Modern Persian," *Translating Esotericism*, eds Wouter J. Hanegraaff and Mriganka Mukhopadhyay, special issue of *Correspondences* 11/1 (2023 [2024]), pp. 103–12.
- Melvin-Koushki, Matthew, "Timurid-Mughal Philosopher-Kings as Sultan-Scientists," *Rulers as Authors in the Islamic World: Knowledge, Authority and Legitimacy*, eds Maribel Fierro, Sonja Brentjes and Tilman Seidensticker, Leiden: Brill, 2024, pp. 596–630.

- Melvin-Koushki, Matthew, "Qizilbash Magic: An Edition of 'Ali Safi b. Husayn Va'iz Kashifi's *Boon for Majd al-Din*, Exemplar of Timurid-Safavid Sunni-Shi'i Occultism," *Iran Namag* 6/3–4 (2021 [2023]), pp. 282–326.
- Melvin-Koushki, Matthew, "Shaykhi Prayer as Safavid-Qajar Occult Technology," English introduction to Muṣṭafā b. Muḥammad Hādī Khū'ī (d. 1839), *Hidden Gems and Treasured Pearls (Javāhir-i maknūna u la'ālī-yi makhzūna)*, eds Alireza Asghari and Muhammad Abdullahiyan, Leiden: The Islamic Manuscripts Press, 2022, pp. 5–31.
- Melvin-Koushki, Matthew, "Dr. Dee's Ottoman Adventure," *Hellebore* 6 (Samhain 2021), pp. 70–79.
- Melvin-Koushki, Matthew, "Divining Past, Present and Future in the Sand: A Persian-Turkish-Arabic Geomantic Miscellany, ca. Sixteenth Century," *The Ottoman World: A Cultural History Reader, 1400–1700*, eds Hakan T. Karateke and Helga Anetshofer, Berkeley: University of California Press, 2021, pp. 244–53.
- Melvin-Koushki, Matthew, "Toward a Neopythagorean Historiography: Kemālpaşa-zāde's (d. 1534) Lettrist Call for the Conquest of Cairo and the Development of Ottoman Occult-Scientific Imperialism," *Islamicate Occult Sciences in Theory and Practice*, eds Liana Saif, Francesca Leoni, Matthew Melvin-Koushki and Farouk Yahya, Leiden: Brill, 2021, pp. 380–419.
- Melvin-Koushki, Matthew, "Is (Islamic) Occult Science Science?" *Theology and Science* 18/2 (2020), pp. 303–24.
- Melvin-Koushki, Matthew, "World as (Arabic) Text: Mīr Dāmād and the Neopythagoreanization of Philosophy in Safavid Iran," *Studia Islamica* 114/3 (2019), pp. 378–431.
- Melvin-Koushki, Matthew, "Imperial Talismanic Love: Ibn Turka's *Debate of Feast and Fight* (1426) as Philosophical Romance and Lettrist Mirror for Timurid Princes," *Der Islam* 96/1 (2019), pp. 42–86.
- Melvin-Koushki, Matthew, "How to Rule the World: Occult-Scientific Manuals of the Early Modern Persian Cosmopolis," *Journal of Persianate Studies* 11/2 (2018), pp. 140–54.
- Melvin-Koushki, Matthew, "Taḥqīq vs. Taqlīd in the Renaissances of Western Early Modernity," *Philological Encounters* 3/1 (2018), pp. 193–249.
- Melvin-Koushki, Matthew, "Early Modern Islamicate Empire: New Forms of Religiopolitical Legitimacy," *The Wiley-Blackwell History of Islam*, eds Armando Salvatore et al., Hoboken: Wiley-Blackwell, 2018, pp. 353–75.
- Melvin-Koushki, Matthew, "Persianate Geomancy from Ṭūsī to the Millennium: A Preliminary Survey," *Occult Sciences in Pre-modern Islamic Cultures*, eds Nader El-Bizri and Eva Orthmann, Beirut: Orient-Institut Beirut, 2018, 151–99.
- Melvin-Koushki, Matthew, "In Defense of Geomancy: Šaraf al-Dīn Yazdī Rebuts Ibn Ḥaldūn's Critique of the Occult Sciences," *Islamicate Occultism: New Perspectives*, eds Matthew Melvin-Koushki and Noah Gardiner, special double issue of *Arabica* 64/3–4 (2017), pp. 346–403.

- Melvin-Koushki, Matthew, "Powers of One: The Mathematicalization of the Occult Sciences in the High Persianate Tradition," *Intellectual History of the Islamicate World* 5/1 (2017), pp. 127–99.
- Melvin-Koushki, Matthew, "Of Islamic Grammatology: Ibn Turka's Lettrist Metaphysics of Light," *Al-Uṣūr al-Wuṣṭā* 24 (2016), pp. 46–117.
- Melvin-Koushki, Matthew, "The Quest for a Universal Science: The Occult Philosophy of Ṣā'in al-Dīn Turka Iṣfahānī (1369–1432) and Intellectual Millenarianism in Early Timurid Iran," PhD diss., Yale University, 2012.
- Melvin-Koushki, Matthew and Alexander Cummins, "Geomancing is Googling: Modern Earth Divination and the Spirits and Ethics of Tech," *Postmedieval* 16/2 (2025), forthcoming.
- Melvin-Koushki, Matthew and James Pickett, "Mobilizing Magic: Occultism in Central Asia and the Continuity of High Persianate Culture under Russian Rule," *Studia Islamica* 111/2 (2016), pp. 231–84.
- Mīr Dāmād, *Jazawāt u mavāqīt*, ed. 'Alī Awjabī, Tehran: Mīrās-i Maktūb, 1380 Sh./2001.
- Mir-Kasimov, Orkhan, *Words of Power: Ḥurūfī Teachings between Shi'ism and Sufism in Medieval Islam: The Original Doctrine of Faḍl Allāh Astarābādī*, London: I. B. Tauris/Institute of Ismaili Studies, 2015.
- Moin, A. Azfar, *The Millennial Sovereign: Sacred Kingship and Sainthood in Islam*, New York: Columbia University Press, 2012.
- Moin, A. Azfar (ed.), "Sulh-i Kull as an Oath of Peace: Mughal Political Theology in History, Theory and Comparison," special issue of *Modern Asian Studies* 56/3 (2022), pp. 715–1052.
- Morgan, Daniel J., "Spokesman of the Unseen World: Shāh Walī Allāh (1703–62), Islamic Reform and Applied Cosmology in Late-Mughal Delhi," PhD diss., University of Chicago, 2021.
- Nasr, Seyyed Hossein, *Islamic Art and Spirituality*, Albany: SUNY Press, 1987.
- Necipoğlu, Gülru, "Transregional Connections: Architecture and the Construction of Early Modern Islamic Empires," *Reflections on Mughal Art & Culture*, ed. Roda Ahluwalia, New Delhi: The K.R. Cama Oriental Institute; Niyogi Books, 2021, pp. 255–85.
- Necipoğlu, Gülru, "The Scrutinizing Gaze in the Aesthetics of Islamic Visual Cultures: Sight, Insight, and Desire," *Muqarnas* 32 (2015), pp. 23–61.
- Noble, Michael-Sebastian, *Philosophising the Occult: Avicennan Psychology and 'The Hidden Secret' of Fakhr al-Dīn al-Rāzī*, Berlin: de Gruyter, 2021.
- O'Kane, Bernard, *Timurid Architecture in Khurasan*, Costa Mesa, Cal.: Mazda, 1987.
- Orthmann, Eva, "Lettrism and Magic in an Early Mughal Text: Muḥammad Ghawth's *Kitāb al-Jawāhir al-Khams*," *The Occult Sciences in Pre-modern Islamic Cultures*, eds Nader El-Bizri and Eva Orthmann, Beirut: Orient-Institut Beirut, 2018, pp. 223–47.
- Orthmann, Eva, "Court Culture and Cosmology in the Mughal Empire: Humāyūn and the Foundations of the *dīn-i ilāhī*," *Court Cultures in the Muslim World: Seventeenth*

- to *Nineteenth Centuries*, eds Albrecht Fuess and Jan-Peter Hartung, London: Routledge, 2014, pp. 202–20.
- Quinn, Sholeh A., *Historical Writing during the Reign of Shah 'Abbas: Ideology, Imitation and Legitimacy in Safavid Chronicles*, Salt Lake City: University of Utah Press, 2000.
- Page, D. L., *Further Greek Epigrams*, Cambridge: Cambridge University Press, 1981.
- Parodi, Laura E., "'The Distilled Essence of the Timurid Spirit': Some Observations on the Taj Mahal," *East and West* 50 (2000), pp. 535–42.
- Peirce, Charles Sanders, "The Basis of Pragmatism in the Normative Sciences," *The Essential Peirce: Selected Philosophical Writings, Volume 2 (1893–1913)*, ed. The Peirce Edition Project, Bloomington: Indiana University Press, 1998, pp. 371–97.
- Piątak, Łukasz, "Between Philosophy, Mysticism and Magic: A Critical Edition of Occult Writings of and Attributed to Shihāb al-Dīn al-Suhrawardī (1156–1191)," PhD diss., University of Warsaw, 2018.
- Pourjavady, Reza and Sabine Schmidtke, "An Eastern Renaissance? Greek Philosophy under the Safavids (16th–18th Centuries AD)," *Intellectual History of the Islamicate World* 3 (2015), pp. 248–90.
- Radin, Dean, *The Science of Magic: How the Mind Weaves the Fabric of Reality*, New York: Harmony, 2025.
- Radin, Dean, *Real Magic: Ancient Wisdom, Modern Science, and a Guide to the Secret Power of the Universe*, New York: Harmony, 2018.
- Ragep, F. Jamil, "Copernicus and His Islamic Predecessors: Some Historical Remarks," *History of Science* 45 (2007), pp. 65–81.
- Ragep, F. Jamil, "Freeing Astronomy from Philosophy: An Aspect of Islamic Cultural Influence on Science," *Osiris* 16 n.s. (2001), pp. 49–71.
- Rizvi, Sajjad H., "Mullā 'Alī Nūrī," *Philosophy in Qajar Iran*, ed. Reza Pourjavady, Leiden: Brill, 2019, pp. 125–78.
- Rizvi, Sajjad H., "Philosophy as a Way of Life in the World of Islam: Applying Hadot to the Study of Mullā Ṣadrā Shīrāzī (d. 1635)," *Bulletin for the School of Oriental and African Studies* 75/1 (2012), pp. 33–45.
- Rizvi, Sajjad H., "Mīr Dāmād in India: Islamic Philosophical Traditions and the Problem of Creation," *Journal of the American Oriental Society* 131/1 (2011), pp. 9–23.
- Rowland, I. D., "Athanasius Kircher, Giordano Bruno, and the *Panspermia* of the Infinite Universe," *Athanasius Kircher: The Last Man Who Knew Everything*, ed. Paula Findlen, New York: Routledge, 2004, pp. 191–205.
- Roxburgh, David J., *Prefacing the Image: The Writing of Art History in Sixteenth-Century Iran*, Leiden: Brill, 2001.
- Saliba, George, *Islamic Science and the Making of the European Renaissance*, Cambridge, Mass.: MIT Press, 2007.
- Saliba, George, "A Sixteenth-Century Arabic Critique of Ptolemaic Astronomy: The Work of Shams al-Dīn al-Khafri," *Journal for the History of Astronomy* 25 (1994), pp. 15–38.

- Schofield, Katherine Butler, "Music, Art and Power in 'Adil Shahi Bijapur, c. 1570–1630," *Scent upon a Southern Breeze: The Synaesthetic Arts of the Deccan*, ed. Kavita Singh, Mumbai: Marg, 2018, pp. 69–87.
- Şen, Ahmet Tunç, *Forgotten Experts: Astrologers, Science and Authority in the Ottoman Empire, 1450–1600*, Stanford: Stanford University Press, 2025.
- Şen, Ahmet Tunç, "Practicing Astral Magic in Sixteenth-Century Istanbul: A Treatise on Talismans Attributed to Ibn Kemāl (d. 1534)," *Characterizing Astrology in the Pre-Modern Islamic World*, eds Shandra Lamaute and Elizabeth Sartell, special issue of *Magic, Ritual, and Witchcraft* 12/1 (2017), pp. 66–88.
- Şen, Ahmet Tunç, "Reading the Stars at the Ottoman Court: Bāyezīd II (r. 886/1481–918/1512) and His Celestial Interests," *Islamicate Occultism: New Perspectives*, eds Matthew Melvin-Koushki and Noah Gardiner, special double issue of *Arabica* 64/3–4 (2017), pp. 557–608.
- Şen, Ahmet Tunç, "Astrology in the Service of the Empire: Knowledge, Prognostication, and Politics at the Ottoman Court, 1450s–1550s," PhD diss., University of Chicago, 2016.
- Shaw, Gregory. *Hellenic Tantra: The Theurgic Platonism of Iamblichus*. New York: Angelico Press, 2024.
- Sheffield, Daniel, *Cosmopolitan Zarathustras: Religion, Translation, and Prophethood in Iran and South Asia*, forthcoming.
- Sheffield, Daniel, "Exercises in Peace: Āzar Kayvānī Universalism and Comparison in the School of Doctrines," *Sulh-i Kull as an Oath of Peace: Mughal Political Theology in History, Theory and Comparison*, ed. A. Azfar Moin, *Modern Asian Studies* 56/3 (2022), pp. 959–92.
- Sheffield, Daniel, "The Lord of the Planetary Court: Revisiting a 'Nativist Prophet' of Early Modern Iran," *Near and Middle Eastern Studies at the Institute for Advanced Study, Princeton: 1935–2018*, ed. Sabine Schmidtke, Piscataway: Gorgias Press, 2018, pp. 455–66.
- Siniosoglou, Niketas, "Sect and Utopia in Shifting Empires: Plethon, Elissaios, Bedreddin," *Byzantine and Modern Greek Studies* 36/1 (2012), pp. 38–55.
- Struck, Peter T., *Divination and Human Nature: A Cognitive History of Intuition in Classical Antiquity*, Princeton: Princeton University Press, 2016.
- Subtelny, Maria E., "Kāshifī's *Asrār-i qāsimī*: A Late Timurid Manual of the Occult Sciences and Its Safavid Afterlife," *Islamicate Occult Sciences in Theory and Practice*, eds Liana Saif, Francesca Leoni, Matthew Melvin-Koushki and Farouk Yahya, Leiden: Brill, 2021, pp. 267–313.
- Subtelny, Maria E., *Le monde est un jardin: aspects de l'histoire culturelle de l'Iran médiéval*, Paris: Association pour l'avancement des études iraniennes, 2002.
- Taneja, Anand Vivek, *Jinnealogy: Time, Islam, and Ecological Thought in the Medieval Ruins of Delhi*, Stanford: Stanford University Press, 2018.

- Terrier, Mathieu, "La figure de Pythagore comme maître d'ésotérisme et de théologie monothéiste dans la philosophie islamique du XI^e/XVII^e siècle," *Appropriations monothéistes de figures "païennes,"* eds C. Macris, J. Soler and A. Van den Kerchove, special issue of *Revue de l'histoire des religions* 236/4 (2019), pp. 731–66.
- Terrier, Mathieu, "The Wisdom of God and the Tragedy of History: The Concept of Appearance (*badā*) in Mīr Dāmād's Lantern of Brightness," *Philosophy and the Intellectual Life in Shī'ah Islam*, eds S.N. Ahmad and S.H. Rizvi, London: The Shī'ah Institute Press, 2017, 94–133.
- Thorndike, Lynn, *A History of Magic and Experimental Science*, 8 vols, New York: Columbia University Press, 1923–58.
- Truschke, Audrey, "Translating the Solar Cosmology of Sacred Kingship," *The Medieval History Journal* 19/1 (2016), pp. 136–41.
- Uy, Cyril V. II, "Riffing on/as the Real: Towards Jazz as a Framework for Medieval Sufism," forthcoming.
- Uy, Cyril V. II, *Lost in a Sea of Letters: Sa'd al-Dīn Ḥamūya and the Plurality of Sufi Knowledge*, Leiden: Brill, 2025.
- Vallée, Jacques, "A Theory of Everything (Else): Another Physics for 2061," TEDx-Brussels, 22 November 2011, <http://www.tedxbrussels.eu/2011>.
- van Bladel, Kevin, "The Iranian Characteristics and Forged Greek Attributions in the Arabic *Sirr al-asrār* (*Secret of Secrets*)," *Mélanges de l'Université Saint-Joseph* 57 (2004), pp. 151–72.
- Vedral, Vlatko, *Decoding Reality: The Universe as Quantum Information*, Oxford: Oxford University Press, 2010.
- Vidal, Gore, *Julian: A Novel*, Boston: Little, Brown, 1964.
- von Stuckrad, Kocku, "Rewriting the Book of Nature: Kabbalah and the Metaphors of Contemporary Life Sciences," *Journal for the Study of Religion, Nature and Culture* 2/4 (2008), pp. 419–42.
- Walbridge, John, *The Wisdom of the Mystic East: Suhrawardī and Platonic Orientalism*, Albany: SUNY Press, 2001.
- Walbridge, John, *The Leaven of the Ancients: Suhrawardī and the Heritage of the Greeks*, Albany: SUNY Press, 2000.
- Wasserstrom, Steven, "Sefer Yešira and Early Islam: A Reappraisal," *Journal of Jewish Thought and Philosophy* 3 (1993), pp. 1–30.
- Westman, Robert S., *The Copernican Question: Prognostication, Skepticism, and Celestial Order*, Berkeley: University of California Press, 2011.
- Wheeler, John Archibald, "Information, Physics, Quantum: The Search for Links," *Complexity, Entropy, and the Physics of Information*, ed. W. H. Zurek, Redwood City, Cal.: Addison-Wesley, 1990, pp. 3–28.
- Wiederhold, Lutz, "Some Remarks on Mālikī Judges in Mamluk Egypt and Syria," *Die Mamlūken: Studien zu ihrer Geschichte und Kultur. Zum Gedenken an Ulrich*

- Haarmann (1942–1999)*, eds Stephan Conermann and Anja Pistor-Hatam, Hamburg: EB Verlag, 2003, pp. 403–13.
- Woods, John E., *The Aqquyunlu: Clan, Confederation, Empire*, rev. ed., Salt Lake City: University of Utah Press, 1999.
- Yazdī, Sharaf al-Dīn ‘Alī, *Ẓafarnāma*, eds Sa‘īd Mīr-Muḥammad Ṣādiq and ‘Abd al-Ḥusayn Navā‘ī, 2 vols, Tehran: Kitābkhāna-yi Mūza u Markaz-i Asnād-i Majlis-i Shūrā-yi Islāmī, 1387 Sh./2008.
- Zadeh, Travis, *Wonders and Rarities: The Marvelous Book That Traveled the World and Mapped the Cosmos*, Cambridge, Mass.: Harvard University Press, 2023.